# FINAL REPORT OF SUMMER INTERNSHIP

# AT ASHOK LEYLAND LIMITED, HOSUR - I

Submitted By


**RAGUL S J**

**23R231**


**BACHELOR OF ENGINEERING**

Specializing in

**ROBOTICS AND AUTOMATION ENGINEERING**


**DEPARTMENT OF ROBOTICS AND AUTOMATION**

**PSG COLLEGE OF TECHNOLOGY (AUTONOMOUS)**

**PEELAMEDU, COIMBATORE– 641004**


**JULY – 2026**

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**INTERNSHIP PERIOD: 11$^{th}$ MAY 2026 – 03$^{rd}$ JULY 2026**

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# DECLARATION

I hereby declare that the internship report entitled "Final Report of Summer Internship at Ashok Leyland Limited, Hosur – I" is an original work carried out and prepared by me, **Ragul S J (Reg. No. 23R231)**, a student of Bachelor of Engineering in Robotics and Automation Engineering, PSG College of Technology, Coimbatore.

This report is based on the knowledge, observations, practical exposure, and learning gained during my internship at Ashok Leyland Limited, Hosur – I. The contents of this report are original and have been prepared by me based on the work carried out and the guidance received during the internship period.

To the best of my knowledge, this report has not been copied from any other source and has not been submitted, either wholly or partly, to any other institution or university for the award of any degree, diploma, certificate, or qualification.

I further declare that all information presented in this report is true and accurate to the best of my knowledge and belief.

**Place:** Coimbatore

**Date:** 03/07/2026

**Ragul S J**
**(23R231)**
Robotics and Automation Engineering,
PSG College of Technology,
Coimbatore – 641 004.

# ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Ashok Leyland Limited, Hosur – I, for providing me with the opportunity to undergo my Summer Internship and gain valuable industrial exposure.

I extend my heartfelt thanks to **Mr. G. Ramesh**, Head – Human Resources, for permitting and supporting my internship at Ashok Leyland. I would like to sincerely thank **Ms. Poonam Bandari**, Human Resources, for her continuous guidance, coordination, and support throughout the internship period.

I would like to extend my heartfelt thanks to **Mr. Ramesh Krishnan K, Maintenance Manager**, for guiding me throughout my internship project. His expert guidance, constructive suggestions, and continuous support were instrumental in the successful completion of my project work and in enhancing my understanding of industrial automation, maintenance engineering, and manufacturing practices.

I express my sincere gratitude to **Mr. Malarvasagan K, Divisional Manager (ALRECON)**, for his valuable guidance, encouragement, and technical support throughout the internship. His mentorship helped me understand industrial quality practices and enhanced my practical knowledge.

I would also like to express my sincere gratitude to **Mr. Sasikumar, Team Leader (Leadec - Maintenance)**, for his continuous support during the internship. He helped me understand the project requirements, coordinated project activities on the shop floor, and provided valuable assistance throughout the project.

I would also like to thank all the employees, supervisors, engineers, and staff members of Ashok Leyland who shared their knowledge and experiences during my internship. Their cooperation and encouragement made this learning experience both meaningful and memorable.

Finally, I express my sincere gratitude to the faculty members and the Department of Robotics and Automation, PSG College of Technology, for their guidance and support in enabling me to successfully complete this internship.



**Ragul S J**

**(23R231)**

Robotics and Automation Engineering,

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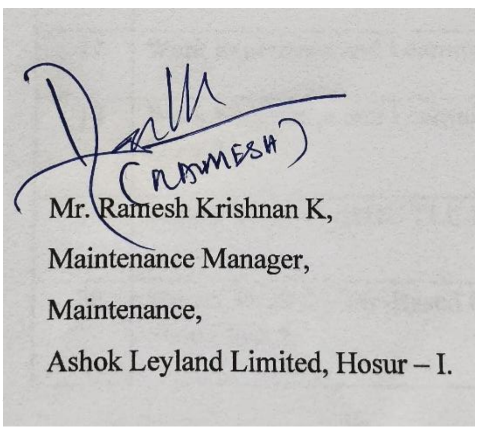
# APPROVAL

This internship report entitled "Final Report of Summer Internship at Ashok Leyland Limited, Hosur – I" submitted by Ragul S J (Reg. No. 23R231), a student of B.E. Robotics and Automation Engineering, PSG College of Technology, Coimbatore, is based on the work carried out during the Summer Internship at Ashok Leyland Limited, Hosur – I. The report has been reviewed and verified to the best of our knowledge and is approved as the Record of the internship work carried out under our guidance.

Industrial Mentor Signature


(RAMESH)

Mr. Ramesh Krishnan K,

Maintenance Manager,

Maintenance,

Ashok Leyland Limited, Hosur – I.

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## HINDUJA GROUP AND ITS ASSOCIATION WITH ASHOK LEYLAND

The Hinduja Group is one of India's largest and most diversified multinational business conglomerates, with a history spanning over a century. Founded in 1914 by Shri Parmanand Deepchand Hinduja, the Group began as a trading enterprise and gradually expanded into a global organization operating in more than 100 countries across six continents. Headquartered in Mumbai, the Group has established a strong presence in various industries through its commitment to innovation, entrepreneurship, ethical business practices, and social responsibility.

The Group's operations cover a broad spectrum of sectors, including automotive manufacturing, banking and financial services, healthcare, information technology, energy, infrastructure, media, real estate, and international trade. This diversified portfolio has enabled the Hinduja Group to maintain sustainable growth while adapting to changing market demands. Guided by its philosophy, "My Duty is to Work So That I Can Give," the Group emphasizes value creation for both business and society through responsible corporate practices and community development initiatives.

Among the many companies within the Hinduja Group, Ashok Leyland is one of its flagship enterprises. The acquisition of Ashok Leyland in 1987 marked a significant milestone in the company's growth journey. With the strategic support of the Hinduja Group, Ashok Leyland modernized its operations, strengthened its manufacturing capabilities, and enhanced its competitiveness within the automotive industry.

The Group's continuous investments in technology, research and development, quality systems, and advanced manufacturing have enabled Ashok Leyland to develop innovative products that meet evolving customer requirements and global standards. This support has helped the company expand its product portfolio across commercial vehicles, buses, defense vehicles, and emerging mobility solutions.

Under the leadership of the Hinduja Group, Ashok Leyland has expanded its presence both within India and in international markets across Asia, Africa, and the Middle East. Today, it is recognized as one of India's leading commercial vehicle manufacturers and a major player in the global bus and transportation sector. The company continues to focus on innovation, sustainability, electrification, and alternative fuel technologies to address future mobility needs.

The Hinduja Group has also contributed significantly to fostering a culture of operational excellence, employee development, workplace safety, and continuous improvement within Ashok Leyland. Through various initiatives in skill development, leadership enhancement, and quality management, the Group has helped build a strong foundation for long-term organizational growth.

The partnership between the Hinduja Group and Ashok Leyland stands as a successful example of visionary leadership, technological advancement, and sustainable business development. Together, they continue to contribute significantly to India's industrial growth while creating value for customers, employees, shareholders, and society.

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# ABOUT ASHOK LEYLAND

Ashok Leyland Limited is one of India's leading commercial vehicle manufacturers and a flagship company of the Hinduja Group. Headquartered in Chennai, Tamil Nadu, the company has been a major contributor to the growth of the Indian automotive industry for over seventy years. Established to meet the country's increasing transportation and mobility demands, Ashok Leyland has built a strong reputation for delivering reliable, efficient, and innovative transportation solutions. Today, it is recognized as the second-largest commercial vehicle manufacturer in India and one of the world's largest bus manufacturers.

The company offers a diverse range of products designed to serve various transportation requirements. Its portfolio includes heavy and medium-duty trucks, light commercial vehicles, buses, defense vehicles, special application vehicles, engines, power solutions, and electric mobility products. These products support industries such as logistics, construction, mining, agriculture, public transportation, defense, and infrastructure development, contributing significantly to economic progress.

Innovation and engineering excellence have always been central to Ashok Leyland's growth. The company invests heavily in research and development to improve vehicle performance, fuel efficiency, safety, durability, and environmental sustainability. Its focus on advanced technologies has led to the development of alternative fuel and electric vehicles, including solutions based on CNG, LNG, hydrogen, and battery-electric technologies, supporting the transition toward sustainable mobility.

Ashok Leyland operates several modern manufacturing facilities across India equipped with advanced production technologies, automation systems, lean manufacturing practices, and rigorous quality control procedures. These facilities enable the company to maintain high production standards while meeting the expectations of customers in both domestic and international markets.

The company has established a significant global presence and exports its products to more than fifty countries across Asia, Africa, the Middle East, Europe, and Latin America. Supported by a strong network of dealers, distributors, and service centres, Ashok Leyland continues to strengthen its position in international markets while maintaining its leadership within India.

In addition to business excellence, the company places great importance on employee welfare, workplace safety, skill development, and ethical business practices. It promotes a culture of teamwork, innovation, and continuous improvement through various training and development initiatives that enhance employee capabilities and organizational performance.

Ashok Leyland is also committed to environmental sustainability and corporate social responsibility. The company undertakes various initiatives related to energy efficiency, waste reduction, water conservation, emission control, and green manufacturing practices. Its CSR activities focus on education, healthcare, road safety, community development, and skill enhancement, creating a positive impact on society.

With a strong legacy and a forward-looking approach, Ashok Leyland continues to embrace digital transformation, Industry 4.0 technologies, connected vehicle systems, and sustainable mobility solutions. Through its commitment to innovation, quality, and customer satisfaction, the company remains a key contributor to India's industrial growth and the global transportation sector.

# HISTORY OF ASHOK LEYLAND

Ashok Leyland has a rich history that reflects its growth into one of India's leading commercial vehicle manufacturers. The company was originally established in 1948 as Ashok Motors in Chennai, primarily to assemble Austin passenger cars. During the early years of India's industrial development, the company contributed to strengthening the domestic automotive sector and supporting the nation's transportation needs.

A major milestone was achieved in 1955 when Ashok Motors entered into a collaboration with Leyland Motors of the United Kingdom, a renowned manufacturer of commercial vehicles. Following this partnership, the company was renamed Ashok Leyland Limited and shifted its focus toward the production of trucks and buses. This marked the beginning of its journey in the commercial vehicle industry.

The collaboration brought modern vehicle technology to India and enabled the company to manufacture reliable transportation solutions for the country's rapidly expanding economy. As industrialization increased, Ashok Leyland's vehicles became widely recognized for their durability, performance, and dependability in both passenger and goods transportation.

During the 1960s and 1970s, the company expanded its operations and introduced a variety of truck and bus models to meet growing market demands. Its buses gained widespread acceptance among public and private transport operators, while its trucks became popular across logistics, industrial, and commercial sectors. These developments helped establish Ashok Leyland as a trusted name in the automotive industry.

To support increasing demand, the company invested in expanding its manufacturing infrastructure and improving product quality. Continuous efforts were made to enhance vehicle efficiency, safety, reliability, and overall performance, enabling the organization to strengthen its competitive position.

A significant transformation occurred in 1987 when the Hinduja Group acquired Ashok Leyland. Under the leadership of the Hinduja Group, the company underwent modernization and strategic expansion. Investments in technology, manufacturing systems, research and development, and workforce development accelerated its growth and strengthened its market presence.

Over the following years, Ashok Leyland focused heavily on innovation and engineering excellence. Dedicated research and development facilities were established to develop advanced vehicle technologies, fuel-efficient engines, emission-compliant products, and improved safety systems. The company also expanded into alternative fuel and electric vehicle technologies to address evolving market and environmental requirements.

Recognizing opportunities in the defense sector, Ashok Leyland diversified its operations by developing specialized vehicles for the Indian Armed Forces. These included logistics vehicles, troop carriers, and other defense mobility solutions, further broadening the company's contribution to national development.

To strengthen its business ecosystem, the company established subsidiaries and strategic partnerships in areas such as vehicle financing, electric mobility, power solutions, aftermarket services, and technology development. These initiatives enabled Ashok Leyland to offer integrated mobility solutions and enhance customer support.

The expansion of manufacturing facilities across locations including Chennai, Hosur, Alwar, Bhandara, and Pantnagar significantly increased production capacity and operational efficiency. Among these, the Hosur manufacturing complex has emerged as one of the company's major production centers, equipped with advanced manufacturing technologies, automation systems, and robust quality assurance practices.

In recent years, Ashok Leyland has embraced digitalization, smart manufacturing, Industry 4.0 technologies, connected vehicles, and sustainable mobility solutions. The company continues to invest in electric vehicles, alternative fuels, and advanced technologies to remain competitive in the rapidly evolving automotive sector.

Today, Ashok Leyland is recognized as a symbol of innovation, quality, and industrial excellence. From its beginnings as a vehicle assembler to becoming a global commercial vehicle manufacturer, the company has demonstrated consistent growth through technological advancement, customer focus, and a commitment to sustainable development. Its continued success reflects its significant contribution to India's transportation industry and economic progress.

# MAJOR MILESTONES OF ASHOK LEYLAND

**1948 –** Ashok Motors Limited was founded in Chennai to assemble Austin passenger cars in India.

**1955 –** Formed a partnership with Leyland Motors, United Kingdom, and adopted the name Ashok Leyland Limited, marking its entry into the commercial vehicle industry.

**1955 –** Manufactured its first commercial vehicle, establishing a foundation in truck and bus production.

**1967 –** Achieved the production milestone of 50,000 vehicles, reflecting its growing presence in the Indian automotive sector.

**1970s –** Expanded manufacturing operations and strengthened its position as a leading commercial vehicle manufacturer in India.

**1980 –** Commissioned the Hosur manufacturing plant, which later became one of the company's major production facilities.

**1987 –** Became part of the Hinduja Group, initiating a new phase of modernization, technological advancement, and business expansion.

**1993 –** Earned ISO 9002 certification, highlighting the company's commitment to quality management and continuous improvement.

**1997 –** Introduced CNG-powered buses, supporting cleaner and more environmentally friendly transportation solutions.

**2002 –** Enhanced its technological capabilities through the introduction of advanced engine technologies and increased focus on research and development.

**2005 –** Celebrated 50 years of association with Leyland Motors and five decades of excellence in commercial vehicle manufacturing.

**2008 –** Expanded into the light commercial vehicle segment, broadening its product portfolio and market reach.

**2010 –** Established the Nissan-Ashok Leyland joint venture for the development and production of light commercial vehicles.

**2012 –** Strengthened its international footprint by expanding operations across various global markets.

**2014 –** Launched the Captain series of trucks, reinforcing its leadership in the heavy commercial vehicle segment.

**2016** – Introduced advanced commercial vehicles designed to meet evolving safety, performance, and emission requirements.

**2017 –** Successfully implemented BS-IV compliant vehicles across its product range.

**2020 –** Transitioned to BS-VI emission standards, demonstrating its commitment to environmental sustainability and technological excellence.

**2021 –** Expanded its contribution to the defense sector through the development and supply of specialized military mobility solutions.

**2022 –** Increased investments in electric mobility and sustainable transportation technologies, including initiatives under Switch Mobility.

**2023**–Present – Continues to focus on digital transformation, Industry 4.0 technologies, electric vehicles, alternative fuels, sustainable manufacturing practices, and global business expansion while maintaining its position as one of India's leading commercial vehicle manufacturers.

## HOSUR UNIT MILESTONES

**1974** – SIPCOT Phase-I industrial land allotted for the Hosur project.

**25 November 1979** – Bhoomi Pooja performed by Mr. R. J. Shahaney

**28 September 1980** – Official inauguration of Ashok Leyland Hosur Unit-I.

**1980s–1990s** – Expansion of manufacturing operations and development of advanced production facilities.

**2000s** – Modernization of production systems with automation and lean manufacturing practices.

**2010s** – Increased focus on engine manufacturing, quality enhancement, and environmental sustainability.

**2020s** – Continued development as a major manufacturing hub supporting commercial vehicles, defense vehicles, and advanced engine production for domestic and international markets.

## VISION, MISSION AND CORE VALUES

**<u>Vision</u>:** To be among the top ten global commercial vehicle manufacturers.

**<u>Mission</u>:** To deliver innovative, reliable, and sustainable transport solutions that exceed customer expectations and generate value for all stakeholders.

**Ashok Leyland is driven by core values such as:**

Ashok Leyland's organizational culture is built upon the following core values:

- Customer centricity
- Innovation
- Integrity
- Sustainability
- Operational excellence

**Overview of Manufacturing Operations**

During my internship at Ashok Leyland, Hosur, I had the opportunity to observe various manufacturing and assembly operations carried out within the plant. The Hosur facility is one of the company's major production centers and plays a significant role in the manufacturing of engines used in different commercial vehicle platforms. The plant integrates modern manufacturing practices with quality-focused operations to ensure efficient production and customer satisfaction.

The facility utilizes automated equipment, assembly stations, testing systems, and quality assurance processes to maintain consistent product standards. Throughout the manufacturing process, emphasis is placed on safety, productivity, process discipline, and continuous improvement. The exposure provided valuable insights into large-scale industrial manufacturing and production management.

One of the notable aspects of the Hosur plant is its dedicated engine manufacturing division, where multiple engine variants are assembled and tested before being dispatched for vehicle integration. The major engine platforms manufactured at the facility include the P15 Series, H-Series, and ZD30 Series engines.

- **<u>P15 Series</u>**
- **<u>H-Series</u>**
- **<u>ZD30 series</u>**

**<u>P15 Engine</u>**

The P15 engine is primarily used in Ashok Leyland's light commercial vehicles such as DOST and BADA DOST. It is designed to provide improved fuel efficiency, dependable performance, and compliance with current emission regulations. The production process is divided into two modules to streamline assembly activities and improve operational efficiency.

**P15 Module I**

In Module I, major engine components are assembled according to standardized work procedures. The assembly process follows a systematic sequence, ensuring that all components meet specified quality requirements before moving to subsequent stages.

**P15 Module II – Women Assembly Line**

During the internship, I observed the P15 Module II assembly line, which is operated entirely by women employees. The team performs assembly, inspection, testing, and quality verification activities with a high level of efficiency and precision. This initiative demonstrates the company's commitment to creating an inclusive work environment and promoting equal opportunities within manufacturing operations.

**H-Series Engine**

The H-Series engine is developed for medium and heavy commercial vehicles. These engines are designed to deliver higher power output and durability for demanding applications such as long-haul transportation, logistics, and construction operations. The manufacturing and testing processes are carried out with strict quality controls to ensure reliable performance under various operating conditions.

**ZD30 Engine**

The ZD30 engine is another important product manufactured at the Hosur facility. Known for its robust construction and dependable operation, the engine is used in specific commercial vehicle applications. The production of the ZD30 engine highlights the plant's capability to handle multiple engine platforms while maintaining high manufacturing standards.

The exposure gained from observing these manufacturing operations provided valuable practical knowledge regarding engine assembly processes, production planning, quality assurance practices, workplace organization, and modern manufacturing systems followed in the automotive industry.

## KEY PRODUCTS MANUFACTURED BY ASHOK LEYLAND

Ashok Leyland offers a diverse range of products designed to meet the transportation needs of various industries. The company's product portfolio includes commercial vehicles, buses, defense vehicles, engines, power solutions, and electric mobility products. These products serve sectors such as logistics, public transportation, construction, mining, agriculture, and defense.

***1. Trucks***

Ashok Leyland manufactures a wide range of trucks for goods transportation, including:

| |
|---|
| Light Commercial Trucks |
| Intermediate Commercial Trucks |
| Heavy Commercial Trucks |
| Tipper Trucks |
| Tractor Trailers |
| Multi-Axle Vehicles |

These vehicles are widely used in logistics, construction, mining, and industrial applications.

***2. Light Commercial Vehicles (LCVs)***

The company offers light commercial vehicles designed for last-mile connectivity and cargo transportation.

Major Models:

| |
|---|
| DOST |
| BADA DOST |
| PARTNER |
| MiTR |

These vehicles are known for fuel efficiency, reliability, and low operating costs.

***3. Buses***

Ashok Leyland is one of the world's largest bus manufacturers and provides buses for various applications:

| |
|---|
| City Buses |
| School Buses |
| Staff Transportation Buses |
| Intercity Buses |

| |
|---|
| Luxury Coaches |

These buses are widely operated by state transport corporations and private fleet operators.

***4. Defense Vehicles***

Ashok Leyland supplies specialized vehicles to the defense sector, including:

| |
|---|
| Troop Carriers |
| Logistics Vehicles |
| Armored Platforms |
| Specialized Military Transport Vehicles |

These vehicles support the mobility requirements of the Indian Armed Forces.

***5. Engines and Power Solutions***

The company manufactures engines for commercial vehicles and industrial applications.

| Major Engine Platforms: |
|---|
| P15 Series Engine |
| H-Series Engine |
| ZD30 Engine |

Ashok Leyland also provides diesel generator engines and other power solutions.

***6. Electric Mobility Solutions***

Through its electric mobility initiatives, the company develops environmentally friendly transportation solutions such as:

| |
|---|
| Electric Buses |
| Electric Light Commercial Vehicles |
| Battery-Based Mobility Solutions |

These products support sustainable transportation and reduced emissions.

### *7. Aftermarket and Service Solutions*

In addition to vehicles, Ashok Leyland offers:

| |
|---|
| Spare Parts |
| Maintenance Services |
| Fleet Management Solutions |
| Vehicle Service Networks |
| Digital Support Services |

These services help improve vehicle uptime and customer satisfaction. The diversified product portfolio of Ashok Leyland enables the company to cater to a wide range of transportation requirements while maintaining its position as one of India's leading commercial vehicle manufacturers.

# POLICIES AT ASHOK LEYLAND

| Policy | Purpose | Key Focus Areas |
|---|---|---|
| **Quality Management Policy** | Ensures customer satisfaction through high-quality products and services. | Continuous improvement, defect prevention, process excellence, innovation, quality assurance. |
| **Sustainability Policy** | Promotes environmentally responsible and socially sustainable business practices. | Energy conservation, emission reduction, resource efficiency, circular economy, CSR activities. |
| **EHS Policy** | Maintains a safe workplace and minimizes environmental impact. | Workplace safety, occupational health, pollution prevention, resource conservation, safety training. |
| **Whistleblower Policy** | Provides a secure channel for reporting unethical practices. | Confidentiality, protection from retaliation, fair investigation, ethical conduct. |
| **POSH Policy** | Ensures a respectful and harassment-free work environment. | Prevention of harassment, grievance redressal, employee awareness, fair investigation. |

# LEAN WORKPLACE ORGANIZATION THROUGH 5S

## OVERVIEW

Ashok Leyland adopts the 5S methodology as a fundamental workplace management tool to improve productivity, safety, quality, and operational efficiency. The initiative promotes a clean, organized, and disciplined work environment, enabling employees to perform their tasks effectively while minimizing waste and unnecessary activities. The implementation of 5S supports the company's commitment to Lean Manufacturing and continuous improvement.

## IMPLEMENTATION OF 5S:

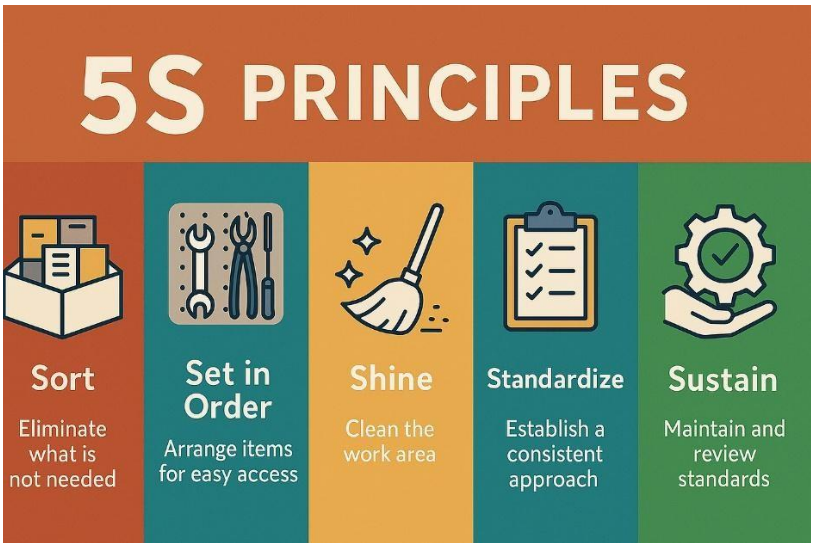
**Sort (Seiri):** Identify and remove all unnecessary items from the workspace. Keep only what is essential for the task at hand**.**

**Set in Order (Seiton):** Organize the remaining items logically so they are easy to find, use, and return. Give everything a clearly labeled, designated home.

**Shine (Seiso):** Clean the workspace and equipment thoroughly. Regular cleaning acts as a visual inspection to spot leaks, damage, or wear and tear.

**Standardize (Seiketsu):** Create consistent rules, schedules, and visual cues for the first three steps. This ensures that everyone knows exactly how the workplace should look and function.

**Sustain (Shitsuke):** Build the discipline to maintain the 5S standards daily. It turns organization into an ongoing habit rather than a one-time clean-up effort.



## ALPS: Enhancing Quality and Productivity

ALPS is Ashok Leyland's manufacturing excellence framework designed to improve productivity, quality, safety, cost efficiency, and delivery performance. It integrates Lean Manufacturing principles and continuous improvement practices to eliminate waste and enhance operational efficiency across all production processes.

**Key Focus Areas of ALPS:**

| |
|---|
| Safety First |
| Quality at Source |
| Cost Reduction |
| On-Time Delivery |
| Employee Involvement |
| Continuous Improvement (Kaizen) |

**Benefits of ALPS:**

| |
|---|
| Improved productivity |
| Reduced process waste |
| Enhanced product quality |
| Better workplace safety |
| Increased customer satisfaction |

**"ALPS drives operational excellence by promoting a culture of quality, efficiency, and continuous improvement throughout the organization."**

## INDUCTION PROGRAM AND DEPARTMENT VISITS

As part of the internship induction program at Ashok Leyland, an orientation schedule was organized to familiarize interns with the company's various departments and manufacturing operations. The visits provided valuable insights into the functions, responsibilities, and interdepartmental coordination involved in vehicle manufacturing.

The departments visited during the induction program included:

**Quality Control Department** – Understanding quality inspection procedures, defect identification, and quality assurance practices followed in production.

**Plant Engineering (Maintenance) – Shop V** – Learning about equipment maintenance strategies, preventive maintenance, and machine reliability.

**P-15 MVC Shop and Assembly** – Observing manufacturing and assembly operations involved in vehicle production.

**Utilities Department** – Studying the utility systems that support plant operations, including power, compressed air, and water management.

**MVC Shop V** – Gaining exposure to production processes, workflow management, and shop-floor activities.

**Engine Assembly Line V** – Understanding engine assembly procedures, component integration, and quality checks.

**Tool Room Shop III and LTMS** – Learning about tool manufacturing, maintenance, calibration, and support provided to production departments.

These departmental visits helped develop a comprehensive understanding of Ashok Leyland's manufacturing environment, operational practices, safety standards, and quality-focused culture. The induction program served as a foundation for understanding the overall production system and prepared interns for their project work within the organization.

# ALLOTMENT FORM

ASHOK LEYLAND

Date: 11/05/2026

| PPSl.No | Department | Person to Meet | HOD |
|---|---|---|---|
| 1 | Quality Control | M/s Srinivasan / Karthick | Mr.Gokulakishnan D |
| 2 | Plant Engg (Maintenance) SHOP V | M/s Joyi | Mr Sabin |
| 3 | P-15 M/C Shop & Assy | M/s Vipin / Sundaravadivel | Mr Vinod Koshy Skaria |
| 4 | Utilities | M/s Rajendra Kumar KVN | Mr C L Subramanya |
| 5 | M/C Shop V | M/s Muralidharan / Vignesh S | Mr N Rajasekaran |
| 6 | Engine Assy II & V | M/s Bernad Joseph / Ramkumar | Mr Mahendran T |
| 7 | Tool Room Shop-III and LTMS | M/s Bashkar / Shubham Kukreti | Mr.J Singaravelu |

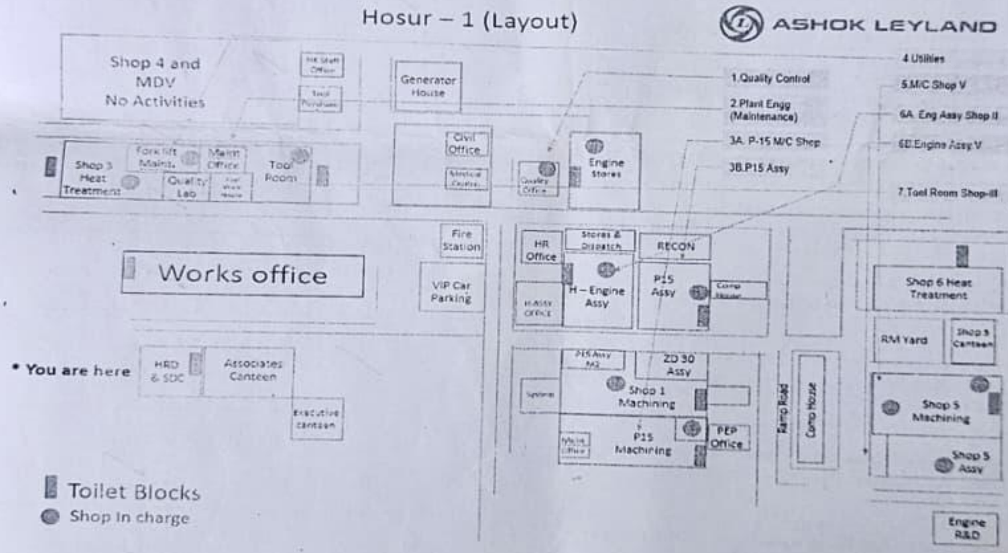
**Lunch: 11.45 -12.15 pm**

**Dear Sir,**

Sub: Training Schedule for [Category] from 2026-05-11 to 2026-07-03

| S.No | Name | Qualification | Academic Year | College Name | Referred By |
|---|---|---|---|---|---|
| 1. | Ragul S J | BE. Robotics and Automation | [Academic Year] | PSG College of Technology | Poonam |

Hosur – 1 (Layout) ASHOK LEYLAND




| Plant Introduction | G. Ramesh Div Mgr – HR Ph: +91 8012580523 | For security Use | Mentor details | Canteen fee Status : |
|---|---|---|---|---|
| S.J. Ragul student signature | | Noted By: | Name: K. Malar Vasagam Area: Recon | Laptop Permission: Make/Brand: S.No: |

- Keep this letter safe until your period gets completed
- No Mobile phones permitted
- Follow Safety rules

# WORK EXPERIENCE AND LEARNINGS IN THE ALRECON DEPARTMENT

## INTRODUCTION TO THE ALRECON DEPARTMENT

The ALRECON (Ashok Leyland Reconstruction) Department is a specialized division responsible for the remanufacturing of used diesel engines that have completed their operational service life or warranty period. Instead of manufacturing engines entirely from new raw materials, the department focuses on recovering reusable components from returned engines, restoring them through standardized engineering processes, and assembling them into fully functional remanufactured engines that meet the quality and performance standards of a new engine. This approach significantly contributes towards sustainable manufacturing by reducing material consumption, minimizing industrial waste, conserving natural resources, and lowering manufacturing costs without compromising product quality and reliability.

The remanufacturing process followed in the department is highly systematic and involves several sequential operations such as engine receipt, cleaning, dismantling, inspection, dimensional verification, machining, component recovery, engine assembly, performance testing, and final dispatch. Every stage is governed by strict quality control procedures to ensure that only components meeting the specified engineering standards are reused in the rebuilt engine. Components that fail to satisfy dimensional or visual inspection criteria are rejected and replaced with new parts, thereby ensuring that the final product delivers reliable performance under actual operating conditions.

Working in the ALRECON department provided extensive exposure to practical manufacturing and production engineering practices. The internship offered an opportunity to observe the complete life cycle of engine remanufacturing and understand how mechanical engineering principles, quality assurance techniques, production planning, inventory management, industrial safety, and lean manufacturing concepts are integrated into a single production system. It also demonstrated how modern industries successfully combine technical expertise with sustainable manufacturing practices to achieve higher productivity while reducing environmental impact.

Another significant aspect observed during the internship was the importance of standardization and traceability. Every engine entering the department is assigned a unique identification number, enabling complete tracking of the engine throughout the remanufacturing process. From dismantling to dispatch, each operation is documented and verified to ensure complete process traceability and compliance with quality standards. Such systematic documentation not only improves production control but also facilitates troubleshooting and quality auditing whenever required. Apart from technical learning, the internship also provided valuable exposure to industrial work culture, teamwork, communication among different functional departments, and systematic problem-solving approaches adopted in a large manufacturing organization. The knowledge gained during the internship established a strong connection between theoretical concepts learned during undergraduate studies and their practical implementation in an industrial environment. The following sections describe the complete remanufacturing activities carried out in the ALRECON department, the engineering practices adopted during engine rebuilding, quality assurance procedures, inventory management systems, technical observations, and the major learning outcomes obtained during the internship.

## ENGINE REMANUFACTURING PROCESS IN ALRECON

The engine remanufacturing process followed in the ALRECON department is a structured sequence of operations aimed at restoring used engines to a condition that closely resembles a newly manufactured engine in terms of performance, reliability, and operational life. Every engine passes through multiple stages of inspection, cleaning, machining, assembly, and testing before it is approved for dispatch. Each stage plays a critical role in ensuring that the remanufactured engine satisfies Ashok Leyland's quality standards.

## RECEIPT OF USED ENGINES AND INITIAL INSPECTION

The remanufacturing process begins with the receipt of engines returned from customers after prolonged field operation. These engines are generally received after completion of their warranty period or when major overhauling becomes necessary. Since the operating conditions vary considerably from one vehicle to another, every engine arrives with different levels of wear, contamination, and component damage. Immediately after arrival, the engine undergoes an initial visual inspection to identify external defects, missing components, oil leakage, damaged housings, cracks, corrosion, and signs of abnormal operation. This inspection helps engineers estimate the overall condition of the engine and determine whether it is suitable for remanufacturing. The engine is then assigned a unique RECON serial number, which serves as the primary identification throughout the rebuilding process. This identification enables complete traceability of the engine and ensures that every process carried out on the engine can be monitored and documented efficiently.

## DEGREASING AND CLEANING OPERATIONS

Since engines returned from field service contain large quantities of grease, lubricating oil, carbon deposits, dirt, and other contaminants, cleaning forms one of the most important preliminary operations in the remanufacturing process. Initially, the engine undergoes a degreasing operation where accumulated oil deposits and grease are removed using specialized cleaning equipment. This process improves the visibility of the engine surfaces and allows accurate inspection of cracks, wear marks, and damaged areas that may otherwise remain hidden beneath contaminants. Following degreasing, the engine components are subjected to high-pressure industrial RO water washing. The use of Reverse Osmosis water minimizes mineral deposits and prevents corrosion during the cleaning process. High-pressure water effectively removes carbon particles, machining chips, dust, and residual contaminants from engine components.

Smaller components such as covers, timing plates, manifolds, flywheel housings, and similar lightweight parts are also cleaned individually before being transferred to the inspection section. Proper cleaning ensures that subsequent inspection and machining operations can be carried out with greater accuracy. During the internship, it was observed that lightweight components often became unstable during high-pressure washing due to the force exerted by the water jet. This occasionally increased handling effort for the operators and indicated opportunities for introducing suitable holding fixtures to improve operator safety and process efficiency.

**COMPONENT RECOVERY AND INSPECTION**

After cleaning, the engine is dismantled into individual components, allowing each part to be inspected separately. The recovered components include the cylinder block, cylinder head, crankshaft, camshaft, intake manifold, exhaust manifold, timing case, flywheel housing, pistons, connecting rods, bearings, covers, and several auxiliary parts. Each component undergoes detailed visual inspection to identify defects such as cracks, corrosion, excessive wear, deformation, damaged threads, broken mounting points, or surface damage. Components that satisfy the required visual standards are forwarded for dimensional inspection, while severely damaged components are rejected immediately.

Visual inspection is one of the most critical stages because the reliability of the remanufactured engine depends directly on the quality of the recovered components. The inspection process requires significant operator experience since several defects can only be identified through careful observation and engineering judgement. During the internship, the cylinder block inspection process was observed in detail. Operators manually examined every surface using high-intensity lighting to identify defects that could affect engine performance after remanufacturing. This inspection demanded a high level of concentration and attention to detail because even minor cracks or defects could lead to engine failure during operation.

## DIMENSIONAL INSPECTION AND QUALITY VERIFICATION

Following the visual inspection, all reusable components are subjected to dimensional inspection to verify whether they satisfy the specified engineering tolerances. This stage plays a vital role in ensuring that only components capable of delivering reliable performance are reused during engine assembly. Various precision measuring instruments such as Vernier calipers, outside micrometers, inside micrometers, bore gauges, dial indicators, depth gauges, and feeler gauges are used for dimensional verification. Measurements including cylinder bore diameter, crankshaft journal diameter, camshaft bearing diameter, piston clearance, connecting rod alignment, valve guide clearance, and several other critical dimensions are carefully inspected according to the manufacturer's specifications.

The measured values are compared with the standard tolerance limits specified by Ashok Leyland. Components falling within the acceptable range are approved for further processing, while those exceeding the permissible limits are either reconditioned through machining operations or replaced with new components. This systematic inspection minimizes the possibility of premature engine failure after remanufacturing. The dimensional inspection process also ensures interchangeability of components during assembly and contributes significantly towards maintaining product quality consistency. Maintaining dimensional accuracy is essential for achieving proper lubrication, efficient combustion, reduced vibration, and longer engine life.

## MACHINING AND RECONDITIONING OPERATIONS

After dimensional inspection, approved components requiring restoration undergo various machining operations to recover their original dimensions and functional characteristics. The objective of these machining processes is to eliminate wear caused during engine operation and restore the component to a condition suitable for reuse. Depending upon the condition of the component, several machining operations are carried out. These include cylinder boring, cylinder honing, crankshaft grinding, cylinder head resurfacing, valve seat cutting, valve grinding, thread restoration, polishing, and surface finishing operations.

Cylinder honing is particularly important because it produces the required cross-hatch surface pattern inside the cylinder liner, which improves lubrication retention and assists in proper piston ring seating during engine operation. Similarly, crankshaft grinding restores the journal surfaces to maintain proper bearing clearances and reduce frictional losses. During machining, every component is continuously monitored to ensure dimensional accuracy and surface finish. After completion of machining, the components are cleaned once again to remove metallic chips, abrasive particles, and machining residues before proceeding to assembly. These machining operations significantly extend the service life of engine components and reduce the overall cost of remanufacturing while maintaining the performance standards expected from a rebuilt engine.

## ENGINE ASSEMBLY PROCESS

Once all components successfully complete the inspection and machining stages, the engine assembly process begins. Assembly is carried out in a sequential manner following standard operating procedures established by Ashok Leyland to ensure consistency and quality. The cylinder block forms the foundation of the engine assembly. Bearings are first installed, followed by careful positioning of the crankshaft. Bearing clearances are verified before tightening the bearing caps according to the recommended torque specifications. Proper torque application is extremely important because excessive tightening may damage the bearings, whereas insufficient tightening may result in component loosening during engine operation.

After crankshaft installation, pistons together with connecting rods are assembled into their respective cylinders. Piston rings are checked for correct orientation and end gap before installation. The camshaft, timing gears, oil pump, lubrication components, cylinder head, rocker arm assembly, valves, manifolds, fuel injection components, water pump, flywheel housing, timing cover, and remaining auxiliary systems are assembled progressively according to the standard assembly sequence. Throughout the assembly process, emphasis is placed on maintaining cleanliness. Even minute dust particles or metallic contaminants can adversely affect engine performance and durability. Therefore, all components are thoroughly cleaned before installation, and lubricating oil is applied wherever necessary to facilitate smooth initial operation. Every fastener is tightened using calibrated torque wrenches to ensure uniform tightening and compliance with engineering specifications. This systematic assembly procedure ensures that the rebuilt engine possesses the structural integrity and operational reliability expected from a quality remanufactured product.

## DRESSING AND PRE-TEST INSPECTION

Following completion of engine assembly, the engine undergoes dressing operations. Dressing refers to the installation of all external accessories and supporting components required before performance testing. Components such as fuel lines, coolant pipes, electrical wiring, sensors, brackets, filters, pulleys, belts, hoses, mounting brackets, and protective covers are fitted carefully. All fluid connections are checked to ensure leak-free operation. Before transferring the engine to the testing section, a comprehensive inspection is carried out. Fasteners are rechecked, lubrication oil is filled, coolant passages are inspected, and manual rotation of the crankshaft is performed to verify smooth mechanical movement. This inspection ensures that the engine is ready for performance evaluation without the risk of mechanical damage.

## ENGINE PERFORMANCE TESTING

Performance testing is one of the most critical stages in the entire remanufacturing process. Every rebuilt engine is tested under controlled operating conditions before being approved for dispatch. The ALRECON department utilizes a SAJ Engine Dynamometer to evaluate engine performance. The dynamometer simulates actual vehicle operating conditions by applying controlled mechanical loads to the engine while continuously monitoring important operating parameters. During testing, parameters such as engine speed (RPM), torque, brake horsepower, fuel consumption, lubricating oil pressure, coolant temperature, exhaust temperature, smoke levels, vibration, and abnormal noise are closely monitored. The engine is operated at different speed and load conditions to evaluate its overall performance. The testing process also helps identify oil leakage, coolant leakage, abnormal combustion, excessive vibration, unusual mechanical sounds, or improper fuel injection characteristics. If any abnormality is detected, the engine is removed from the test bench, the root cause is identified, corrective actions are performed, and the engine undergoes retesting until all specified performance standards are achieved. Only engines successfully passing all performance tests receive final quality approval. This comprehensive testing procedure ensures that customers receive engines capable of delivering reliable service under actual operating conditions.

## QUALITY ASSURANCE AND INVENTORY MANAGEMENT

Quality assurance forms an integral part of every activity performed within the ALRECON department. Rather than inspecting quality only at the final stage, quality control measures are implemented throughout the complete remanufacturing process. Inspection checkpoints are established after cleaning, dismantling, machining, assembly, and performance testing to ensure that defects are identified at the earliest possible stage. Standard operating procedures, inspection checklists, calibrated measuring instruments, and documented quality records are extensively used to maintain consistency. Traceability of every engine is maintained using unique identification numbers, allowing the manufacturing history of each remanufactured engine to be tracked whenever required. Inventory management also plays a significant role in ensuring uninterrupted production. Reusable components, newly procured spare parts, fasteners, seals, bearings, gaskets, and consumables are systematically organized to facilitate quick identification and retrieval. Proper inventory control minimizes production delays, prevents shortages, and improves overall operational efficiency. The internship provided valuable exposure to inventory management practices, material traceability systems, and production planning methods adopted in a large-scale automotive manufacturing organization.

## ENGINEERING OBSERVATIONS AND SCOPE FOR IMPROVEMENT

Several engineering practices adopted within the ALRECON department demonstrated the organization's strong commitment towards quality, productivity, and continuous improvement. The systematic workflow, standardized operating procedures, and emphasis on quality assurance significantly contributed towards maintaining high remanufacturing standards. During the observation of washing operations, it was noticed that lightweight engine components occasionally became unstable due to the high-pressure water jet. Introducing dedicated holding fixtures could improve operator convenience, enhance safety, and reduce handling effort during cleaning. Similarly, greater digital integration of inspection records and production monitoring could further strengthen traceability while reducing manual documentation. Expansion of automation in material movement and inspection activities may also improve production efficiency and minimize operator fatigue. The department already follows several lean manufacturing practices, and continuous improvement initiatives can further enhance productivity while maintaining the organization's high-quality standards.

## TECHNICAL KNOWLEDGE AND LEARNING OUTCOMES

The internship in the ALRECON department provided extensive practical exposure to industrial manufacturing processes and significantly enhanced technical understanding in several engineering domains. The major learning outcomes include:

- Understanding the complete engine remanufacturing process from engine receipt to final dispatch.
- Knowledge of engine construction and functioning of major diesel engine components.
- Practical exposure to industrial cleaning, dismantling, machining, inspection, and assembly operations.
- Understanding the application of precision measuring instruments in dimensional inspection.
- Knowledge of quality assurance procedures followed in automotive remanufacturing.
- Understanding the importance of production planning, inventory management, and material traceability.
- Familiarity with engine performance evaluation using a dynamometer.
- Awareness of lean manufacturing concepts and systematic industrial work practices.
- Appreciation of workplace safety, teamwork, documentation, and continuous improvement methodologies.

Overall, the internship served as an excellent platform for bridging theoretical engineering knowledge with real industrial applications. The experience improved technical competence, problem-solving ability, observation skills, and understanding of large-scale automotive manufacturing operations. It also strengthened professional confidence and provided valuable insight into the practical implementation of production engineering concepts within a leading automobile manufacturing organization.

# WORK EXPERIENCE AND LEARNINGS IN THE MAINTENANCE TEAM – SHOP FLOOR 2

## INTRODUCTION

Maintenance plays a vital role in every manufacturing industry by ensuring the continuous, safe, and efficient operation of production equipment. In modern automobile manufacturing industries such as Ashok Leyland, maintenance is not merely confined to repairing failed machines but also encompasses preventive maintenance, predictive maintenance, condition monitoring, equipment improvement, automation support, safety enhancement, and continuous process optimization. A well-established maintenance system minimizes machine downtime, improves equipment reliability, enhances product quality, and contributes significantly towards achieving production targets.

Shop Floor 2 at Ashok Leyland consists of several critical production and testing facilities that require uninterrupted operation throughout the manufacturing process. The maintenance team is responsible for ensuring that every machine, testing system, electrical panel, PLC-controlled equipment, pneumatic circuit, hydraulic unit, conveyor, and utility system functions efficiently under all production conditions. The department performs regular inspections, preventive maintenance schedules, emergency breakdown rectification, root cause analysis, spare parts management, and continuous improvement activities to maximize machine availability.

The internship in the Maintenance Team provided extensive practical exposure to industrial maintenance practices followed in a large-scale automotive manufacturing environment. Unlike conventional academic laboratory experiments, the industrial environment demonstrated how multiple engineering disciplines such as electrical engineering, mechanical engineering, automation engineering, instrumentation, industrial safety, and production engineering work together to maintain uninterrupted manufacturing operations. Every maintenance activity demanded careful planning, systematic troubleshooting, technical expertise, and strict adherence to safety procedures.

The internship provided an opportunity to closely observe the maintenance philosophy adopted by Ashok Leyland, where machine reliability is considered one of the key contributors to production efficiency. It was observed that every maintenance activity was supported by standard operating procedures, technical documentation, preventive maintenance schedules, equipment history records, and systematic fault analysis. This organized approach significantly reduced equipment failures while improving production reliability. Apart from technical knowledge, the internship also enhanced problem-solving skills, logical thinking, teamwork, communication, documentation practices, and the ability to understand industrial equipment from a systems engineering perspective. The experience helped bridge the gap between theoretical engineering concepts and their practical implementation in an industrial manufacturing environment.

## ROLE AND RESPONSIBILITIES OF THE MAINTENANCE TEAM

The Maintenance Team in Shop Floor 2 is responsible for ensuring the continuous availability and reliable operation of all production equipment installed within the shop floor. Since manufacturing operations depend heavily on machine availability, even a minor equipment failure can interrupt production schedules, reduce productivity, and increase manufacturing costs. Therefore, the maintenance department plays an essential role in supporting the overall manufacturing process. One of the primary responsibilities of the maintenance team is the execution of preventive maintenance activities. Machines are periodically inspected according to predetermined maintenance schedules to identify worn-out components before failures occur. Lubrication, alignment checks, tightening of electrical connections, replacement of consumable components, calibration of sensors, verification of safety devices, and inspection of moving mechanisms are performed regularly to maintain equipment in optimum operating condition.

The maintenance team is also responsible for attending breakdown maintenance whenever equipment failures occur unexpectedly. During such situations, technicians and engineers systematically diagnose the fault by analysing electrical circuits, mechanical assemblies, PLC programs, sensors, actuators, pneumatic systems, hydraulic circuits, and communication networks. The objective is not only to restore machine operation quickly but also to identify the actual root cause to prevent recurrence of similar failures. Electrical maintenance forms another important responsibility of the department. Industrial electrical panels containing circuit breakers, contactors, overload relays, variable frequency drives (VFDs), programmable logic controllers (PLCs), relays, power supplies, and communication modules require periodic inspection to ensure reliable operation. The maintenance team regularly monitors electrical parameters, verifies cable connections, checks insulation conditions, and replaces faulty electrical components whenever necessary.

Mechanical maintenance activities involve inspection of bearings, shafts, couplings, gearboxes, chains, sprockets, conveyors, belts, linear guides, rollers, hydraulic cylinders, pneumatic cylinders, pumps, valves, and other rotating equipment. Proper alignment, lubrication, and component replacement are carried out to minimize mechanical failures and improve equipment life. Another significant responsibility involves supporting automation systems installed throughout the shop floor. Modern manufacturing equipment utilizes PLCs, HMIs, industrial communication networks, proximity sensors, photoelectric sensors, pressure transmitters, temperature sensors, flow sensors, and servo drives for automatic operation. Maintenance engineers continuously monitor these systems, troubleshoot communication failures, modify PLC logic whenever required, and ensure proper synchronization between electrical and mechanical systems. The maintenance department also contributes to process improvement by identifying recurring machine failures, analysing downtime records, recommending design modifications, improving equipment accessibility, and implementing engineering solutions that enhance productivity and operator safety.

## INDUSTRIAL MAINTENANCE PRACTICES OBSERVED

One of the major learning experiences during the internship was understanding how maintenance activities are systematically planned rather than performed only after equipment failure. The maintenance philosophy followed in the organization emphasizes reliability, safety, productivity, and continuous improvement. Preventive maintenance schedules are prepared based on equipment operating hours, manufacturer recommendations, and previous maintenance history. Each machine undergoes scheduled inspection at regular intervals to identify potential problems before they develop into major failures. This approach significantly reduces unplanned downtime and improves overall equipment effectiveness.

Condition monitoring techniques are also employed wherever applicable. Instead of replacing components solely based on time intervals, maintenance personnel observe equipment condition through vibration analysis, temperature monitoring, lubrication inspection, visual examination, and operational behaviour. Such practices improve maintenance efficiency while reducing unnecessary replacement of healthy components. During maintenance activities, strict lockout and tagout procedures are followed before working on energized electrical equipment or moving mechanical assemblies. Power isolation, compressed air isolation, hydraulic pressure release, and verification of zero-energy conditions are performed before commencing maintenance work. These safety practices protect maintenance personnel from accidental injuries and ensure compliance with industrial safety standards.

The maintenance department maintains detailed records of equipment history, previous failures, spare parts consumption, inspection reports, maintenance schedules, and corrective actions. These records help engineers identify recurring problems, evaluate equipment reliability, and plan future maintenance strategies more effectively. The internship demonstrated that successful maintenance is not limited to repairing equipment but involves systematic planning, continuous monitoring, effective documentation, engineering analysis, and proactive decision-making. The department's structured maintenance practices significantly contributed towards maintaining uninterrupted production and ensuring high manufacturing efficiency.

## EXPOSURE TO INDUSTRIAL AUTOMATION AND CONTROL SYSTEMS

One of the most valuable aspects of the internship in the Maintenance Team was the opportunity to gain practical exposure to industrial automation systems employed throughout Shop Floor 2. Modern manufacturing industries depend heavily on automation to achieve high productivity, consistent product quality, reduced human intervention, and improved operational safety. During the internship, several automated production systems were studied, providing a comprehensive understanding of how electrical, electronic, mechanical, and software systems function together as an integrated manufacturing solution.

The maintenance department plays a significant role in ensuring the continuous operation of these automation systems. Every automated machine consists of multiple interconnected subsystems such as Programmable Logic Controllers (PLCs), Human Machine Interfaces (HMIs), industrial communication networks, sensors, actuators, electrical control panels, pneumatic systems, and safety devices. A malfunction in any one of these subsystems can interrupt the complete production process. Therefore, maintenance engineers continuously monitor these systems, diagnose faults, replace defective components, and verify proper operation before returning the equipment to production. During the internship, considerable emphasis was placed on understanding the architecture of industrial automation systems rather than merely observing machine operation. The interaction between sensors, PLCs, actuators, motor drives, and operator interfaces was studied in detail, providing practical insight into automated manufacturing processes.

## WORKING WITH PLC-BASED CONTROL SYSTEMS

Programmable Logic Controllers form the heart of modern industrial automation systems. Throughout the internship, extensive exposure was obtained to PLC-controlled machines operating within the shop floor. These controllers continuously receive signals from field sensors, process the information according to the programmed control logic, and generate output commands to operate motors, valves, relays, contactors, solenoid valves, cylinders, pumps, and other industrial equipment. Unlike conventional relay-based control systems, PLCs provide flexibility, reliability, faster response, simplified troubleshooting, and easy modification of control logic. Their ability to handle complex sequential operations makes them indispensable in large manufacturing industries. The maintenance team regularly monitored PLC-controlled equipment to ensure uninterrupted operation. Whenever faults occurred, engineers analysed the PLC input signals, output status, communication modules, alarm messages, and diagnostic indicators before identifying the actual cause of failure. Exposure to PLC programming enhanced the understanding of industrial automation principles and demonstrated how software logic directly controls physical manufacturing equipment.

## UNDERSTANDING INDUSTRIAL ELECTRICAL CONTROL PANELS

Electrical control panels serve as the central control units for industrial machines. During the internship, detailed observation of various electrical panels provided valuable knowledge regarding industrial electrical systems and their organization. A typical industrial control panel consists of incoming power supply sections, miniature circuit breakers (MCBs), moulded case circuit breakers (MCCBs), contactors, overload relays, PLC CPUs, digital and analogue input/output modules, relays, terminal blocks, switched mode power supplies (SMPS), communication modules, fuses, transformers, emergency stop circuits, selector switches, push buttons, and status indication lamps. Each component inside the panel performs a specific function that contributes towards reliable machine operation. Proper cable routing, terminal identification, wire numbering, equipment labelling, and systematic arrangement greatly simplify maintenance and fault diagnosis. During maintenance activities, engineers carefully verified power availability, protection devices, loose electrical connections, damaged cables, faulty relays, defective contactors, blown fuses, and malfunctioning PLC modules before proceeding with detailed troubleshooting. This systematic approach minimized repair time while improving maintenance efficiency. The internship provided practical understanding of electrical panel construction, industrial wiring practices, safety precautions, and systematic electrical fault diagnosis.

## INDUSTRIAL SENSORS AND FIELD DEVICES

Sensors form the primary source of information for every automated manufacturing system. Throughout the internship, different types of industrial sensors installed on production equipment were studied to understand their operating principles and industrial applications. Proximity sensors were widely used for detecting metallic components without physical contact. These sensors enabled automatic positioning, part counting, limit detection, and machine sequencing. Photoelectric sensors were employed wherever non-contact object detection was required irrespective of material type. Pressure switches and pressure transmitters continuously monitored pneumatic and hydraulic systems to ensure that machines operated within safe pressure limits. Temperature sensors were installed on engines, coolant circuits, lubrication systems, and utility equipment to monitor operating temperatures and prevent overheating. Limit switches were used for position feedback in mechanically operated systems, while rotary encoders provided accurate speed and positional information wherever precise motion control was necessary. The internship demonstrated that the overall reliability of an automated manufacturing system depends heavily upon the accurate functioning of these field devices. Even a single faulty sensor can interrupt the complete production cycle. Therefore, maintenance engineers regularly inspected sensor alignment, wiring integrity, mounting conditions, operating range, and electrical outputs during preventive maintenance activities.

## TROUBLESHOOTING METHODOLOGY FOLLOWED BY THE MAINTENANCE TEAM

One of the most important engineering skills acquired during the internship was understanding the structured troubleshooting methodology adopted by experienced maintenance engineers. Whenever a machine failure occurred, maintenance personnel did not immediately replace components. Instead, a logical sequence of fault diagnosis was followed to accurately identify the root cause. Initially, the operator's observations regarding the machine behaviour were collected. Alarm messages displayed on the HMI, PLC diagnostic indicators, fault history records, and machine operating conditions were analysed. Visual inspection was then carried out to identify damaged cables, loose connectors, oil leakage, broken mechanical components, abnormal noise, overheating, or physical damage.

Electrical parameters such as voltage, current, continuity, and insulation condition were verified wherever necessary. Sensor inputs and PLC outputs were monitored to determine whether control signals were reaching the intended devices. Communication between PLC modules and field equipment was also verified. If electrical systems were functioning correctly, attention shifted towards mechanical components such as bearings, couplings, belts, gearboxes, pneumatic cylinders, hydraulic valves, and moving assemblies. This systematic elimination approach enabled maintenance engineers to accurately isolate the defective component without unnecessary replacement of healthy equipment. The troubleshooting methodology observed during the internship highlighted the importance of engineering analysis, logical reasoning, systematic observation, technical documentation, and practical experience in solving industrial maintenance problems efficiently.

## PREVENTIVE AND BREAKDOWN MAINTENANCE ACTIVITIES

Preventive maintenance and breakdown maintenance together form the foundation of industrial maintenance management. Preventive maintenance involves planned inspection and servicing activities carried out before equipment failure occurs. These activities include lubrication of moving components, tightening electrical terminals, inspection of safety devices, cleaning of electrical panels, calibration of sensors, checking pneumatic pressure, verification of hydraulic oil levels, replacement of consumable components, and functional testing of machine operations. Regular preventive maintenance significantly improves equipment reliability while reducing unexpected production interruptions.

In contrast, breakdown maintenance is performed whenever equipment unexpectedly fails during production. Such failures demand immediate response from the maintenance department to minimize production losses. The maintenance team rapidly analyses the fault, identifies the defective component, performs the necessary repairs or replacements, verifies machine operation, and restores production within the shortest possible time. During the internship, both preventive and corrective maintenance activities were observed. The experience demonstrated the importance of proper planning, spare parts availability, teamwork, communication between production and maintenance personnel, and systematic fault analysis in achieving efficient maintenance performance. The maintenance practices followed within Shop Floor 2 clearly illustrated that successful maintenance extends far beyond repairing machines. It involves continuous monitoring, engineering analysis, preventive action, documentation, process improvement, and effective coordination among various departments to ensure uninterrupted manufacturing operations.

## PRACTICAL EXPOSURE TO PLC PROGRAMMING AND INDUSTRIAL CONTROL

One of the most significant learning experiences during the internship was gaining practical exposure to Programmable Logic Controller (PLC) programming and industrial control systems used in Shop Floor 2. Modern manufacturing industries rely heavily on PLC-based automation for controlling machines, monitoring equipment, acquiring process data, and ensuring smooth production operations. The maintenance department plays a crucial role in maintaining these automation systems by diagnosing faults, modifying control logic when required, verifying input and output signals, and ensuring reliable communication between various field devices. During the internship, practical exposure was obtained to Siemens PLC systems programmed using Totally Integrated Automation (TIA) Portal. The software environment provides a comprehensive platform for PLC programming, hardware configuration, Human Machine Interface (HMI) development, communication setup, diagnostics, and system monitoring. Working with TIA Portal helped in understanding how industrial control logic is designed, tested, modified, and implemented within an actual manufacturing environment.

The maintenance activities demonstrated that PLC programming is not limited to writing ladder logic alone. It involves proper hardware configuration, address allocation, selection of suitable data types, program optimization, alarm handling, communication management, signal conditioning, and integration with various industrial devices. Understanding these practical aspects significantly improved knowledge of industrial automation beyond classroom concepts. The interaction between PLC hardware, field sensors, electrical control panels, communication modules, and Human Machine Interfaces illustrated how complex manufacturing operations are coordinated automatically with high accuracy and reliability. Observing these systems under actual production conditions provided valuable practical knowledge regarding industrial automation architecture and real-time machine control.

## UNDERSTANDING PLC HARDWARE CONFIGURATION

An important aspect of industrial automation involves understanding the physical architecture of PLC systems. During the internship, different PLC hardware components were studied to understand their individual functions and their contribution to the complete automation system. The Central Processing Unit (CPU) acts as the brain of the PLC, continuously scanning input signals, executing the programmed logic, and updating output devices. Digital Input modules receive signals from switches, sensors, push buttons, emergency stop circuits, and proximity sensors, whereas Digital Output modules control relays, contactors, solenoid valves, indicator lamps, and various field actuators. Analogue Input modules receive continuously varying process signals such as pressure, temperature, flow rate, and level measurements from transmitters installed throughout the manufacturing system. Similarly, Analogue Output modules generate proportional control signals for devices such as control valves, variable frequency drives, and other process control equipment. Communication modules facilitate information exchange between PLCs, HMIs, remote I/O stations, drives, and supervisory systems through industrial communication protocols. Power supply modules ensure stable operation of all PLC components by providing regulated DC power. Understanding the function of each hardware module greatly simplified the interpretation of PLC programs and assisted in diagnosing hardware-related faults during maintenance activities.

## MONITORING MACHINE PARAMETERS USING PLC

Industrial automation systems continuously monitor numerous machine parameters to ensure safe and efficient operation. Throughout the internship, several machine variables were observed being monitored through PLC-based control systems. These parameters included motor operating status, engine rotational speed, coolant temperature, lubricating oil pressure, compressed air pressure, emergency stop conditions, machine interlocks, process timings, sensor feedback signals, actuator status, alarm conditions, and equipment operating sequences. The PLC continuously processes these signals in real time and makes control decisions based on the programmed logic. If any operating parameter exceeds its permissible limit, the controller immediately generates alarms, initiates protective actions, or safely shuts down the equipment to prevent damage. Understanding the importance of real-time monitoring highlighted how industrial automation improves operational safety, equipment protection, and production efficiency.

## PRACTICAL UNDERSTANDING OF HIGH-SPEED SIGNAL PROCESSING

A particularly interesting area of learning during the internship involved understanding how high-speed pulse signals generated by rotating equipment are processed within industrial automation systems. Unlike ordinary digital inputs that monitor ON and OFF conditions, high-speed pulse signals require specialized processing because pulse frequencies may become extremely high during machine operation. Standard PLC scan cycles may not accurately capture these rapidly changing signals. Therefore, dedicated High-Speed Counter (HSC) technology is employed to ensure accurate pulse acquisition without signal loss. The practical implementation of High-Speed Counters demonstrated how rotational speed measurements are obtained from pulse-generating devices such as encoders and speed sensors. The PLC continuously counts incoming pulses over a specified time interval and converts this information into meaningful engineering parameters such as revolutions per minute (RPM). Understanding this concept provided valuable knowledge regarding industrial motion control, speed monitoring, process measurement, and high-speed data acquisition techniques used extensively in automated manufacturing systems.

## HUMAN MACHINE INTERFACE (HMI) AND OPERATOR INTERACTION

Human Machine Interfaces provide the communication bridge between industrial equipment and machine operators. During the internship, HMI systems were observed being used for machine operation, parameter monitoring, fault diagnosis, alarm acknowledgement, and equipment status visualization. The HMI displays real-time operating information received from the PLC, enabling operators to monitor machine performance continuously. Parameters such as equipment status, motor conditions, sensor values, process variables, alarms, fault messages, and operating sequences are presented in an organized graphical format. Maintenance engineers also utilize HMI screens during troubleshooting activities to identify abnormal operating conditions and verify equipment performance after repairs. Modifications to display parameters, alarm messages, and process information improve operator awareness and facilitate efficient machine operation. The integration between PLCs and HMIs illustrated the importance of user-friendly operator interfaces in modern industrial automation systems.

## ENGINEERING SKILLS DEVELOPED DURING MAINTENANCE ACTIVITIES

Working with the maintenance team contributed significantly towards the development of both technical and professional engineering skills. Unlike classroom experiments where equipment operates under controlled conditions, industrial maintenance requires engineers to solve real-time problems under production constraints while maintaining safety and equipment reliability. The internship enhanced analytical thinking by encouraging systematic fault diagnosis rather than random component replacement. Every maintenance activity required observation, logical reasoning, technical discussion, verification of engineering data, and careful interpretation of machine behaviour before arriving at an appropriate solution. Practical exposure also improved the ability to read electrical schematics, understand PLC input-output addressing, identify industrial hardware components, interpret machine documentation, and communicate effectively with experienced engineers and technicians.

Another important learning outcome was the understanding that successful maintenance depends not only on technical knowledge but also on teamwork, planning, documentation, discipline, safety awareness, and effective communication between production, maintenance, quality, and engineering departments. The experience strengthened confidence in handling industrial automation systems and provided a solid foundation for undertaking more advanced automation projects involving PLC programming, HMI development, industrial networking, and machine control systems.

## ENGINEERING OBSERVATIONS AND INDUSTRIAL BEST PRACTICES

Working alongside the maintenance engineers provided valuable insight into the engineering practices followed in a large-scale automobile manufacturing industry. One of the most important observations was that maintenance is not considered an isolated activity but an integral part of the entire manufacturing process. Every maintenance activity is planned with the objective of maximizing machine availability, maintaining product quality, reducing production downtime, and ensuring operator safety. A systematic workflow was followed whenever maintenance activities were carried out. Before initiating any maintenance work, the operating condition of the equipment was assessed, safety precautions were implemented, relevant documentation was reviewed, and the required tools and spare parts were arranged. Such organized planning minimized maintenance time and prevented unnecessary interruptions to production activities.

Another important observation was the emphasis placed on standardization. Standard operating procedures, electrical drawings, maintenance checklists, machine manuals, PLC documentation, and equipment history records were extensively used throughout maintenance activities. These documents enabled engineers to diagnose faults more efficiently and ensured that maintenance activities were performed consistently irrespective of the personnel involved. The maintenance department also encouraged continuous monitoring of equipment performance. Instead of waiting for machine failures to occur, engineers regularly inspected critical components and process parameters to identify early signs of deterioration. This proactive approach significantly improved equipment reliability while reducing maintenance costs. Teamwork was another major factor contributing to successful maintenance operations. Production operators, maintenance technicians, engineers, quality personnel, and supervisors worked together whenever equipment problems occurred. Effective communication between departments enabled rapid fault diagnosis and minimized production losses.

## CONTINUOUS IMPROVEMENT INITIATIVES

The internship demonstrated that continuous improvement is an essential part of industrial maintenance. Every machine breakdown, recurring fault, or operational difficulty was treated as an opportunity to improve the manufacturing process. Maintenance engineers analysed the root causes of failures and implemented corrective measures to prevent similar occurrences in the future. Several improvement initiatives focused on enhancing equipment reliability, reducing downtime, improving operator safety, simplifying maintenance procedures, and increasing production efficiency. Periodic review of maintenance records enabled engineers to identify frequently failing components and recommend suitable engineering modifications wherever necessary.

Automation also played a significant role in continuous improvement activities. The use of PLC-controlled systems, sensor-based monitoring, HMI diagnostics, and automated alarms reduced manual intervention while improving process consistency. The maintenance department actively supported these automation initiatives by modifying control logic, improving equipment integration, and ensuring reliable operation of automated systems. Another important aspect of continuous improvement involved effective spare parts management. Maintaining adequate inventory levels for frequently used components reduced repair time during emergency breakdowns and improved overall maintenance efficiency. The department also promoted systematic housekeeping, organized workplaces, proper cable routing, equipment identification, documentation control, and preventive maintenance schedules, all of which contributed towards maintaining an efficient and safe working environment.

## RELATIONSHIP BETWEEN MAINTENANCE AND PRODUCTION ENGINEERING

The internship clearly demonstrated the close relationship between maintenance engineering and production engineering. Although production departments are responsible for manufacturing activities, their performance depends heavily upon the reliability and availability of production equipment maintained by the maintenance team. Every production schedule requires machines to operate continuously with minimum interruptions. Effective maintenance directly contributes to higher production output, improved equipment utilization, consistent product quality, reduced manufacturing costs, and timely delivery of finished products. Several concepts studied during the Production Engineering curriculum were found to be directly applicable during the internship. Preventive maintenance, Total Productive Maintenance (TPM), Overall Equipment Effectiveness (OEE), reliability engineering, production planning and control, industrial safety, quality assurance, lean manufacturing, and continuous improvement were all observed in practical industrial applications. The maintenance department therefore serves as one of the fundamental pillars supporting modern manufacturing industries by ensuring that production equipment remains available, reliable, and efficient throughout its operational life.

## PROFESSIONAL SKILLS AND COMPETENCIES DEVELOPED

The internship significantly enhanced both technical knowledge and professional competencies. Working in a real industrial environment provided opportunities to observe experienced engineers solving practical engineering problems using systematic analytical methods. This exposure improved logical reasoning, troubleshooting capability, observation skills, and engineering decision-making. Technical skills developed during the internship include understanding industrial automation systems, PLC-controlled machines, electrical control panels, industrial sensors, preventive maintenance procedures, breakdown maintenance, troubleshooting methodologies, machine documentation, and industrial safety practices. Exposure to these technologies strengthened the ability to understand and analyse modern manufacturing equipment.

The internship also improved interpersonal skills such as communication, teamwork, technical discussions, professional discipline, documentation practices, and time management. Interaction with engineers, technicians, supervisors, and production personnel provided valuable insight into the collaborative nature of industrial engineering projects. Another significant learning outcome was the ability to correlate theoretical engineering concepts with their practical implementation. Subjects such as Manufacturing Technology, Industrial Automation, Production Planning and Control, Maintenance Engineering, Mechatronics, Industrial Instrumentation, Quality Control, Fluid Power Systems, and Electrical Engineering became much easier to understand after observing their practical applications within the manufacturing environment. The experience also strengthened confidence in handling industrial equipment, interpreting engineering drawings, understanding machine operations, and approaching technical problems systematically.

## OVERALL LEARNING OUTCOMES

The internship in the Maintenance Team of Shop Floor 2 provided valuable practical exposure and contributed significantly to the development of technical and professional competencies. The major learning outcomes are summarized below:

- Gained comprehensive knowledge of industrial maintenance practices followed in a large-scale automobile manufacturing industry.
- Understood the importance of preventive maintenance, breakdown maintenance, and predictive maintenance in improving machine reliability and reducing downtime.
- Acquired practical exposure to industrial automation systems and their role in modern manufacturing processes.
- Developed a fundamental understanding of PLC-based control systems and their applications in industrial automation.
- Learned the working principles and applications of Human Machine Interface (HMI) systems for machine monitoring and operator interaction.
- Gained knowledge of industrial electrical control panels, including various electrical and automation components used in machine control.

- Understood the operation and industrial applications of field devices such as proximity sensors, photoelectric sensors, pressure sensors, temperature sensors, and limit switches.
- Learned systematic troubleshooting methodologies for identifying and rectifying electrical, mechanical, and automation-related faults.
- Understood the importance of documentation, maintenance records, equipment history, and standard operating procedures in industrial maintenance management.
- Gained knowledge of industrial safety practices, including lockout-tagout (LOTO) procedures and safe maintenance practices.
- Improved analytical thinking and logical problem-solving abilities through observation of real industrial maintenance activities.
- Enhanced technical understanding of machine operation, equipment reliability, and production support systems.
- Developed the ability to correlate theoretical concepts learned in Production Engineering with their practical industrial applications.
- Improved communication skills, teamwork, professional discipline, and interaction with engineers, technicians, and production personnel.
- Understood the multidisciplinary integration of mechanical, electrical, electronics, instrumentation, and automation engineering in modern manufacturing industries.
- Strengthened practical knowledge in industrial automation, manufacturing systems, production engineering, and maintenance engineering.
- Enhanced confidence in understanding industrial machinery, automation systems, and engineering workflows.
- Developed a strong foundation for future work in industrial automation, robotics, manufacturing systems, intelligent maintenance technologies, and production engineering.
- Successfully bridged the gap between academic learning and practical industrial implementation through hands-on exposure in a professional manufacturing environment.
- Overall, the internship significantly improved technical competence, engineering knowledge, professional confidence, analytical thinking, and practical problem-solving skills, contributing to overall professional development as a Production Engineering student.

# PROJECT WORK 1 – AHU PLC PANEL FOR SHOP FLOOR 5

## INTRODUCTION

As part of the internship in the Maintenance Department at Ashok Leyland Limited, an opportunity was provided to participate in the development of a Programmable Logic Controller (PLC) based control panel for the Air Handling Unit (AHU) installed in Shop Floor 5. The project was undertaken with the objective of improving the reliability, maintainability, and operational convenience of the existing AHU control system by implementing an industrial automation solution using Siemens PLC technology. The project involved understanding the working of the Air Handling Unit, analysing the existing electrical control arrangement, selecting suitable automation hardware, designing the PLC control panel, developing the control logic using Siemens Totally Integrated Automation (TIA) Portal, configuring the Human Machine Interface (HMI), carrying out electrical wiring, and performing testing to verify the correct operation of the system.

Unlike conventional industrial maintenance activities, this project provided direct exposure to the complete engineering cycle of an automation project. Every stage required technical understanding, engineering analysis, component selection, programming, electrical integration, testing, and systematic verification before implementation. The experience provided practical knowledge of how automation projects are planned and executed in an industrial environment.

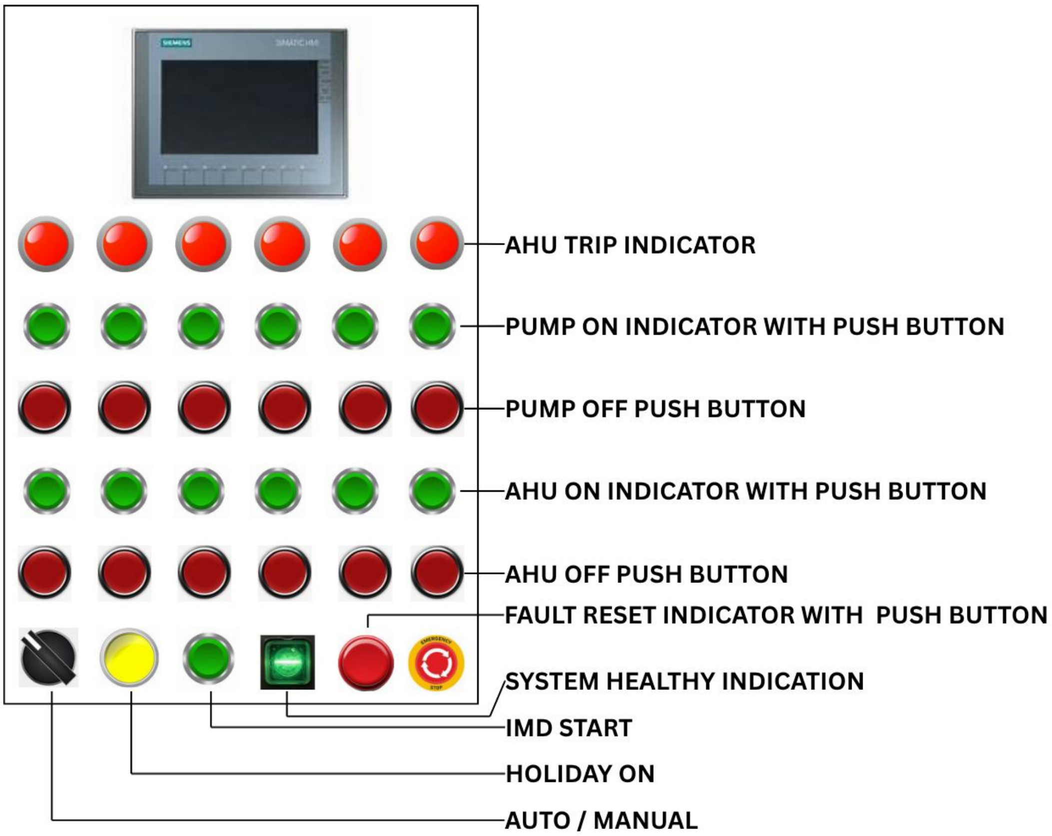



*Figure of the PLC Panel Front Buttons and Indication System*

The Air Handling Unit is an important utility system used within the manufacturing facility to maintain the required environmental conditions inside the shop floor. It ensures proper circulation of air, maintains adequate ventilation, regulates airflow, and contributes to creating a suitable working environment for personnel and manufacturing operations. Reliable operation of the AHU is therefore essential for maintaining comfortable working conditions and ensuring uninterrupted industrial activities.

## PROJECT BACKGROUND

Industrial utility systems such as Air Handling Units are generally expected to operate continuously for long durations with high reliability. Since these systems directly influence the environmental conditions inside the manufacturing shop, any unexpected interruption may affect operator comfort and overall plant operation. The existing control arrangement of the Air Handling Unit was functional but lacked the flexibility and diagnostic capabilities offered by modern PLC-based automation systems. Maintenance activities involving conventional control circuits often require significant effort during troubleshooting because fault identification depends largely on manual inspection of relays, timers, wiring, and control components. In addition, future modifications and expansion of the control system become more difficult when conventional relay logic is employed. Modern industries therefore prefer PLC-based control systems because they simplify troubleshooting, reduce wiring complexity, improve system flexibility, and enable easy modification of operating logic. Considering these advantages, the project focused on developing a dedicated PLC control panel capable of controlling the Air Handling Unit using industrial automation techniques while maintaining reliable and user-friendly operation.

## OBJECTIVES OF THE PROJECT

The primary objective of the project was to design and develop a PLC-based control panel for the Air Handling Unit operating in Shop Floor 5. The automation system was intended to improve operational reliability, simplify maintenance activities, and provide a structured control architecture using Siemens industrial automation components. The specific objectives of the project are listed below:

- To study the construction and working principle of the existing Air Handling Unit.
- To understand the operational requirements of the ventilation system.
- To analyse the existing electrical control arrangement.
- To design a new PLC-based control architecture.
- To select suitable Siemens PLC hardware for the application.
- To prepare the electrical wiring layout for the control panel.
- To develop the PLC control logic using Siemens TIA Portal.

- To configure the Human Machine Interface (HMI) for operator interaction.
- To implement manual speed selection for different operating conditions.
- To verify the functionality of all inputs and outputs.
- To test the complete control panel before implementation.
- To improve system maintainability and simplify future troubleshooting.

## WORKING PRINCIPLE OF THE AIR HANDLING UNIT

An Air Handling Unit (AHU) is a mechanical system used for circulating and distributing conditioned air throughout an industrial building. It forms one of the major components of the Heating, Ventilation, and Air Conditioning (HVAC) system and is responsible for maintaining proper air circulation, ventilation, and indoor air quality within the manufacturing facility. The basic operation of the Air Handling Unit begins with the intake of fresh air from the surrounding environment. Depending on the operating conditions, fresh air may also be mixed with recirculated air returning from the shop floor to improve energy efficiency. The incoming air first passes through a series of filters where dust particles and other contaminants are removed before entering the blower section.

The blower, driven by an electric motor, is the primary component responsible for moving air through the entire system. Once the blower starts operating, air is forced through the duct network and distributed uniformly throughout the shop floor. The continuous circulation of air improves ventilation, maintains suitable working conditions, and assists in removing heat generated during manufacturing operations. In the developed control system, the blower motor operates based on **manual speed selection**. The operator selects the required operating speed using the control interface provided in the PLC-HMI system. Depending upon the selected speed level, the PLC generates the corresponding output signals required for operating the blower at the desired speed. Since the speed is selected manually by the operator, the system does not automatically vary the motor speed according to environmental conditions. The PLC continuously monitors all input conditions and executes the programmed control logic to ensure safe operation of the blower. Before energising the motor, the controller verifies the required operating conditions and then activates the corresponding output sequence. If any abnormal operating condition is detected, the PLC can stop the blower operation and indicate the corresponding status on the Human Machine Interface. The implementation of manual speed selection through PLC control offers improved operational flexibility while maintaining a simple control philosophy. It also enables future expansion of the system if automatic control based on temperature, humidity, or pressure sensors is required.

## OVERALL CONTROL PHILOSOPHY

The control philosophy adopted for this project was intentionally kept simple and reliable to meet the operational requirements of Shop Floor 5. Rather than implementing a fully automatic control strategy, the system was designed around manual operator control using the PLC and HMI. The operator initiates the operation of the Air Handling Unit through the HMI interface. After the start command is received, the PLC verifies the operating conditions and enables the blower motor. The operator can then manually select the desired speed level according to the ventilation requirement of the shop floor. The PLC executes the corresponding logic and activates the appropriate control outputs for the selected operating mode. Throughout operation, the PLC continuously monitors the status of the system and updates the HMI with the current operating condition. Any abnormal status, fault condition, or emergency stop command immediately overrides the running sequence to ensure safe equipment operation. This manual control philosophy provides simplicity, reliability, ease of operation, and straightforward troubleshooting, making it highly suitable for industrial utility applications where operator supervision is readily available.

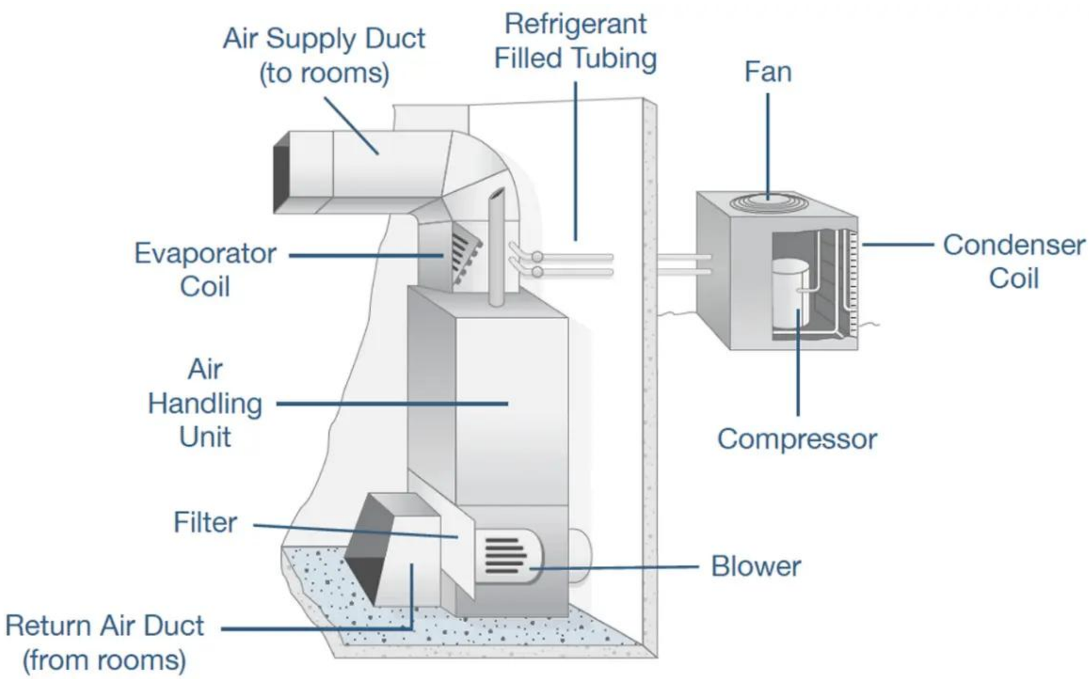



*Figure of the Components of Air Handling Unit*

## SYSTEM REQUIREMENT ANALYSIS

Before designing the PLC control panel, a detailed study of the Air Handling Unit and its operational requirements was carried out. Understanding the existing system was essential for selecting suitable hardware components and developing an appropriate control strategy. The analysis involved identifying the electrical equipment associated with the AHU, understanding the control sequence, studying the available power supply, and determining the required input and output signals for the PLC. The operational requirement of the Air Handling Unit was relatively straightforward, as the blower motor was intended to operate based on manual speed selection. However, the control panel had to ensure safe operation by incorporating suitable protection devices, emergency stop provisions, status indications, and reliable communication between the PLC and the Human Machine Interface (HMI). The system also needed to provide sufficient flexibility for future expansion without requiring major modifications to the control hardware. During the analysis stage, special attention was given to simplifying maintenance activities. Instead of relying entirely on conventional relay logic, the PLC-based system was designed to perform all control functions through programmable logic. This approach not only reduced wiring complexity but also made troubleshooting easier by allowing maintenance engineers to monitor the status of every input and output through the PLC programming software and HMI.

## SELECTION OF PLC HARDWARE

The selection of suitable PLC hardware is one of the most critical stages in the development of any industrial automation system. Since the PLC acts as the central controller, it must possess adequate processing capability, sufficient input and output capacity, reliable communication facilities, and compatibility with future expansion requirements. For this project, a Siemens PLC was selected due to its widespread industrial acceptance, high reliability, ease of programming, excellent diagnostic capabilities, and seamless integration with Siemens Human Machine Interface devices. Siemens PLCs are extensively used in manufacturing industries because of their robust performance, modular architecture, and comprehensive software support through Totally Integrated Automation (TIA) Portal.

The PLC selected for the project was capable of receiving input signals from field devices, executing the programmed control logic, and generating output signals for controlling the blower motor and associated electrical equipment. The CPU continuously scans all input signals, processes the ladder logic program, updates the output status, and communicates with the HMI for displaying machine information. The modular design of the PLC allows additional input and output modules to be incorporated whenever system expansion becomes necessary. This feature makes the control system highly flexible and suitable for future modifications without replacing the existing controller.

## HUMAN MACHINE INTERFACE (HMI)

The Human Machine Interface serves as the communication medium between the operator and the Air Handling Unit control system. Instead of interacting directly with electrical components inside the control panel, the operator can perform all control operations through a graphical display provided by the HMI. The HMI developed for this project provides a user-friendly interface for operating the Air Handling Unit. Through this interface, the operator can initiate system operation, stop the blower whenever required, manually select the desired operating speed, and continuously monitor the running status of the equipment. In addition to operational control, the HMI also provides visual indication of various machine conditions. Running status, stop status, emergency conditions, communication status, and other operational parameters can be displayed clearly, enabling maintenance personnel to quickly understand the current condition of the system.

Another major advantage of incorporating an HMI is the simplification of troubleshooting activities. Instead of tracing electrical wiring manually, maintenance engineers can monitor PLC variables and operating conditions directly through the HMI, thereby reducing diagnostic time and improving maintenance efficiency.

## ELECTRICAL COMPONENTS USED IN THE CONTROL PANEL

The PLC control panel consists of several electrical and automation components that collectively ensure safe and reliable operation of the Air Handling Unit. Each component performs a specific function within the control system. The Miniature Circuit Breaker (MCB) protects the electrical circuits from overload and short-circuit conditions. Whenever excessive current flows through the circuit, the MCB disconnects the supply automatically, preventing damage to electrical equipment. A Switched Mode Power Supply (SMPS) converts the incoming AC supply into regulated 24 V DC power required for operating the PLC, HMI, sensors, and other control devices. Stable DC power is essential for ensuring reliable operation of automation equipment. Industrial relays are employed wherever electrical isolation is required between the PLC outputs and field devices. These relays also provide additional protection to the PLC by preventing high electrical loads from directly reaching the controller outputs.

Contactors are used for switching the blower motor. Since the PLC outputs cannot directly drive high-power motors, the PLC energizes the contactor coil, which in turn connects the main power supply to the motor. Terminal blocks are provided for organized field wiring. They simplify installation, improve maintenance accessibility, and facilitate systematic cable identification. Push buttons and selector switches are incorporated for manual operation, maintenance functions, and emergency shutdown whenever required. Indicator lamps are installed on the control panel to display power availability, machine running condition, fault indication, and communication status, thereby enabling operators to understand the operational condition of the system at a glance.

## PANEL DESIGN AND LAYOUT

The design of the PLC control panel was carried out by considering electrical safety, maintenance accessibility, wiring convenience, and future expandability. Proper arrangement of components inside the panel not only improves aesthetics but also simplifies inspection, troubleshooting, and replacement of electrical devices. The incoming power section was positioned separately from the low-voltage automation section to minimize electrical interference. The PLC, power supply, relays, terminal blocks, communication modules, and other automation devices were mounted systematically on DIN rails with adequate spacing between components for heat dissipation.

Cable ducts were provided for routing all internal wiring in an organized manner. Proper wire numbering and terminal identification were maintained throughout the panel to facilitate future maintenance activities. Labels were also provided for every electrical component, thereby improving identification during troubleshooting. The panel door accommodated the Human Machine Interface, selector switches, push buttons, emergency stop switch, and indication lamps. This arrangement enabled the operator to perform all operational activities without opening the control panel, thereby improving electrical safety. Adequate space was intentionally reserved inside the panel for future addition of communication modules, extra PLC input/output cards, and other automation devices if system expansion becomes necessary.

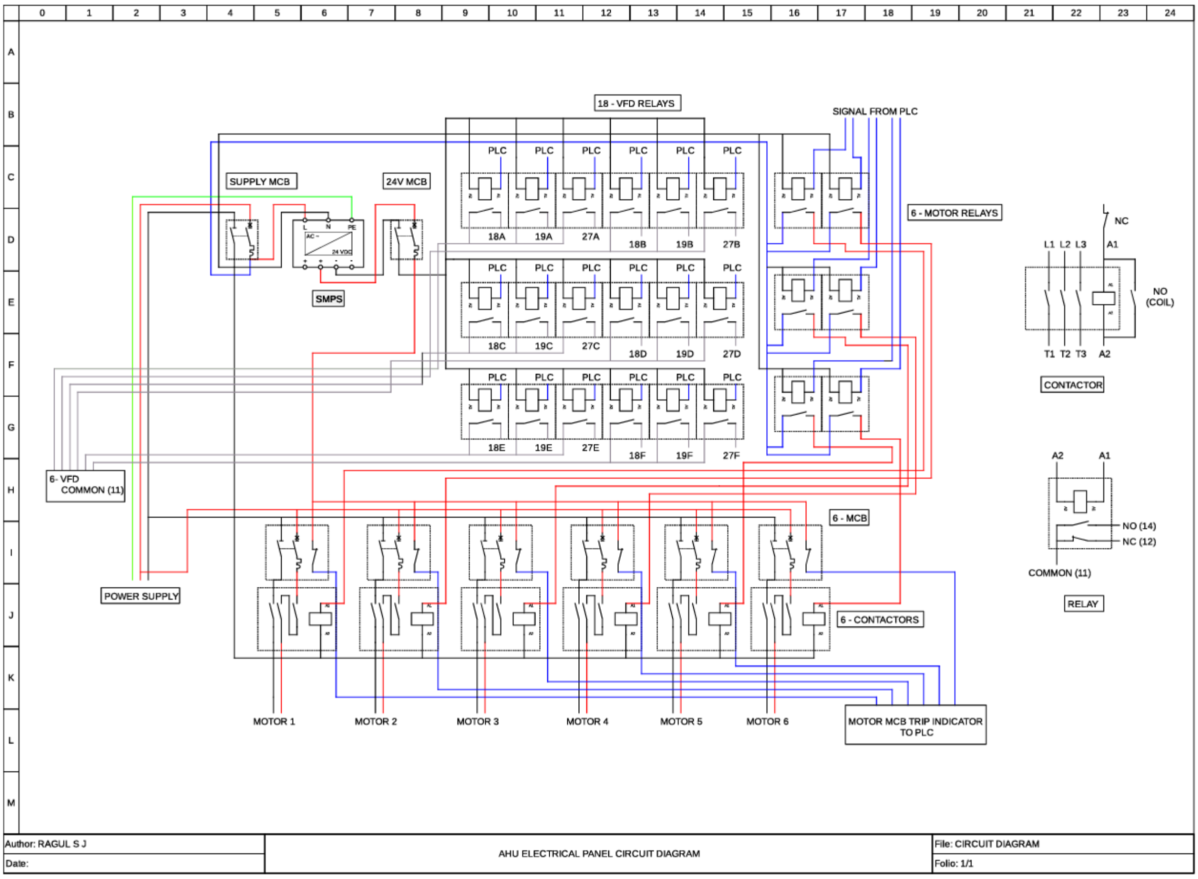



*Figure of the Panel Circuit Diagram*

## PREPARATION OF ELECTRICAL WIRING

After completion of the panel layout, electrical wiring was carried out according to the prepared electrical schematic. Every connection between the PLC, HMI, relays, power supply, contactors, terminal blocks, selector switches, and field devices was made systematically by following standard industrial wiring practices. Colour-coded wires were used to distinguish between AC power circuits, DC control circuits, communication cables, and signal wiring. Ferrules were crimped onto every wire before termination to ensure proper identification and improve long-term reliability.

Special care was taken to maintain proper cable routing inside the cable ducts, avoiding unnecessary crossing of wires. Adequate bending radius was maintained for all conductors, while cable ties were used wherever necessary to improve neatness and mechanical support. After completion of wiring, every electrical connection was verified against the circuit diagram before applying power to the system. Continuity testing and insulation verification were also performed to ensure that no wiring errors existed within the control panel. Proper documentation of all wiring connections was maintained for future maintenance and troubleshooting purposes.

## DEVELOPMENT OF THE PLC CONTROL PROGRAM

The development of the PLC control program formed the core of the Air Handling Unit automation project. After the hardware components were selected and the electrical panel design was completed, the next stage involved developing the control logic that would enable the PLC to operate the Air Handling Unit according to the required operating sequence. The PLC program was developed using Siemens Totally Integrated Automation (TIA) Portal, which provides a comprehensive engineering environment for PLC programming, hardware configuration, communication setup, diagnostics, and Human Machine Interface development. Before beginning the programming process, all hardware modules were configured within the software environment, and the corresponding input and output addresses were assigned according to the electrical wiring diagram.

The entire control program was developed in a structured manner by dividing the control logic into individual functional sections. This modular programming approach simplified program development, debugging, testing, and future modification. Each section of the ladder logic was responsible for a particular operation such as system initialization, start-stop control, manual speed selection, interlocking conditions, output control, and status indication. Special attention was given to writing a simple, reliable, and easily understandable program. Since the system was intended for industrial use, the ladder logic was designed in such a manner that future maintenance engineers could easily interpret the program and perform modifications whenever required.

## PLC INPUT AND OUTPUT CONFIGURATION

A proper input-output configuration is essential for ensuring accurate communication between the PLC and the field devices. Each input device connected to the PLC transmits information regarding the operating condition of the system, while the output devices execute the commands generated by the PLC program. The digital input section consists of field devices such as the Start Push Button, Stop Push Button, Emergency Stop Switch, Auto/Manual Selector (if applicable), and manual speed selection switches. These devices continuously provide the PLC with the necessary information required to execute the programmed control logic.

The digital output section consists primarily of relay outputs responsible for energizing contactors, operating indication lamps, and enabling the selected blower speed. Whenever the PLC receives a valid operating command, it processes the input conditions and energizes the corresponding output channel. Proper addressing of every input and output signal was maintained throughout the programming process. Each address was clearly documented to simplify troubleshooting and future maintenance.

| Input Signal | Address |
|---|---|
| Emergency Stop | I0.0 |
| Auto/Manual | I0.1 |
| Fault Reset | I0.2 |
| Holiday ON | I0.3 |
| Indication of Start | I0.4 |
| Pump 1 ON | I0.5 |
| Pump 2 ON | I0.6 |
| Pump 3 ON | I0.7 |
| Pump 4 ON | I1.0 |
| Pump 5 ON | I1.1 |
| Pump 6 ON | I1.2 |
| Pump 1 OFF | I1.3 |
| Pump 2 OFF | I1.4 |
| Pump 3 OFF | I1.5 |
| Pump 4 OFF | I2.0 |
| Pump 5 OFF | I2.1 |
| Pump 6 OFF | I2.2 |
| AHU 1 ON | I2.3 |
| AHU 2 ON | I2.4 |
| AHU 3 ON | I2.5 |
| AHU 4 ON | I2.6 |
| AHU 5 ON | I2.7 |
| AHU 6 ON | I3.0 |
| AHU 1 OFF | I3.1 |
| AHU 2 OFF | I3.2 |

| Input Signal | Address |
|---|---|
| AHU 3 OFF | I3.3 |
| AHU 4 OFF | I3.4 |
| AHU 5 OFF | I3.5 |
| AHU 6 OFF | I3.6 |
| Pump 1 MCB Trip | I3.7 |
| Pump 2 MCB Trip | I4.0 |
| Pump 3 MCB Trip | I4.1 |
| Pump 4 MCB Trip | I4.2 |
| Pump 5 MCB Trip | I4.3 |
| Pump 6 MCB Trip | I4.4 |
| Booster Pump MCB Trip | I4.5 |
| AHU 1 Trip | I4.6 |
| AHU 2 Trip | I4.7 |
| AHU 3 Trip | I5.0 |
| AHU 4 Trip | I5.1 |
| AHU 5 Trip | I5.2 |
| AHU 6 Trip | I5.3 |

| Output Signal | Address |
|---|---|
| Emergency Stop Indication | Q0.0 |
| Fault Reset Indication | Q0.1 |
| System Healthy | Q0.2 |
| Pump 1 ON Indication | Q0.3 |
| Pump 2 ON Indication | Q0.4 |
| Pump 3 ON Indication | Q0.5 |
| Pump 4 ON Indication | Q0.6 |
| Pump 5 ON Indication | Q0.7 |
| Pump 6 ON Indication | Q1.0 |
| AHU 1 ON Indication | Q1.1 |
| AHU 2 ON Indication | Q1.2 |
| AHU 3 ON Indication | Q2.0 |
| AHU 4 ON Indication | Q2.1 |
| AHU 5 ON Indication | Q2.2 |
| AHU 6 ON Indication | Q2.3 |
| AHU 1 Trip Indication | Q2.4 |
| AHU 2 Trip Indication | Q2.5 |
| AHU 3 Trip Indication | Q2.6 |
| AHU 4 Trip Indication | Q2.7 |
| AHU 5 Trip Indication | Q2.8 |
| AHU 6 Trip Indication | Q3.0 |

| Output Signal | Address |
|---|---|
| Pump 1 Contactor ON | Q3.1 |
| Pump 2 Contactor ON | Q3.2 |
| Pump 3 Contactor ON | Q3.3 |
| Pump 4 Contactor ON | Q3.4 |
| Pump 5 Contactor ON | Q3.5 |
| Pump 6 Contactor ON | Q3.6 |

**MANUAL SPEED SELECTION LOGIC**

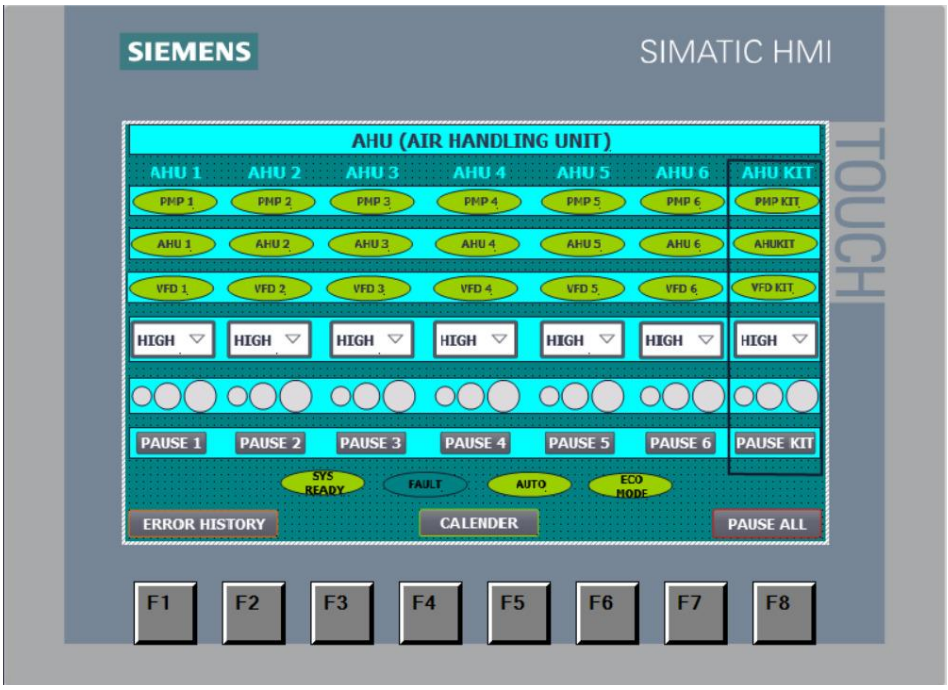
One of the primary objectives of the project was to implement a manual speed selection mechanism for the Air Handling Unit blower motor. Unlike automatic control systems where the speed varies according to process parameters such as temperature or pressure, this system allows the operator to manually select the required blower speed based on the ventilation requirement inside Shop Floor 5. Initially, the operator starts the Air Handling Unit using the designated Start command available through the Human Machine Interface or push button station. After verifying that all operating conditions are satisfied, the PLC energizes the main blower contactor and prepares the system for speed selection. The operator can then select one of the available operating speeds using the speed selection interface. Depending upon the selected option, the PLC activates the corresponding relay output associated with that speed level. To ensure safe operation, the control logic permits only one speed selection to remain active at any given time. Whenever a different speed is selected, the previously active speed is automatically deactivated before energizing the newly selected output.

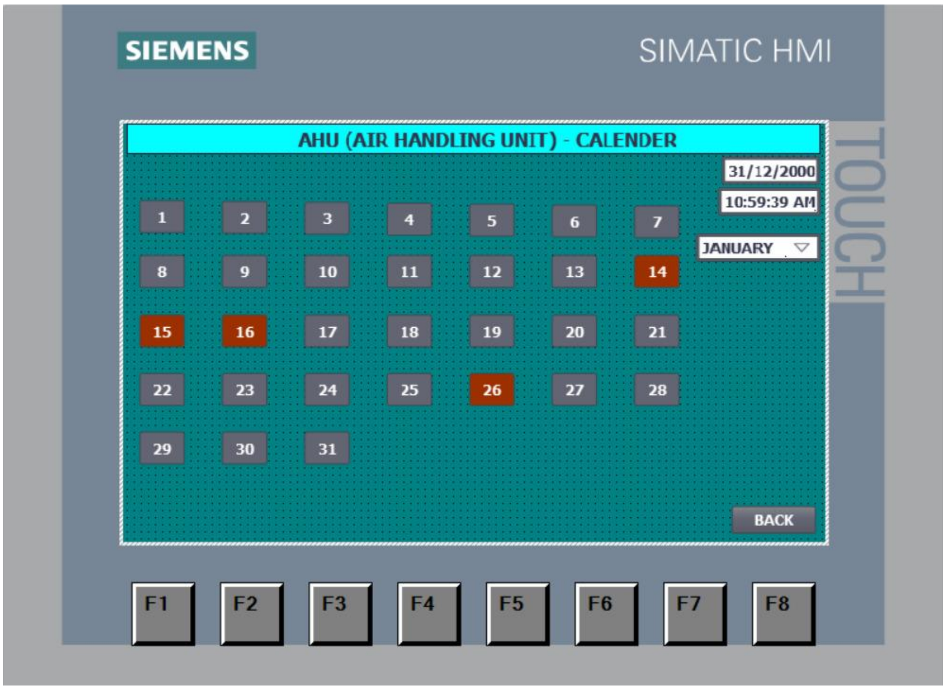
This interlocking arrangement prevents simultaneous activation of multiple speed outputs and eliminates the possibility of conflicting control signals reaching the blower motor. The ladder logic continuously monitors the selected operating mode throughout system operation. Whenever the operator changes the selected speed, the PLC immediately updates the corresponding output while maintaining uninterrupted blower operation. The Stop command immediately de-energizes all output relays and safely shuts down the Air Handling Unit. Similarly, activation of the Emergency Stop circuit overrides every operating command and disconnects all control outputs to ensure maximum operator safety. The implemented control strategy offers simplicity, operational flexibility, and ease of maintenance while providing reliable manual control over the Air Handling Unit.

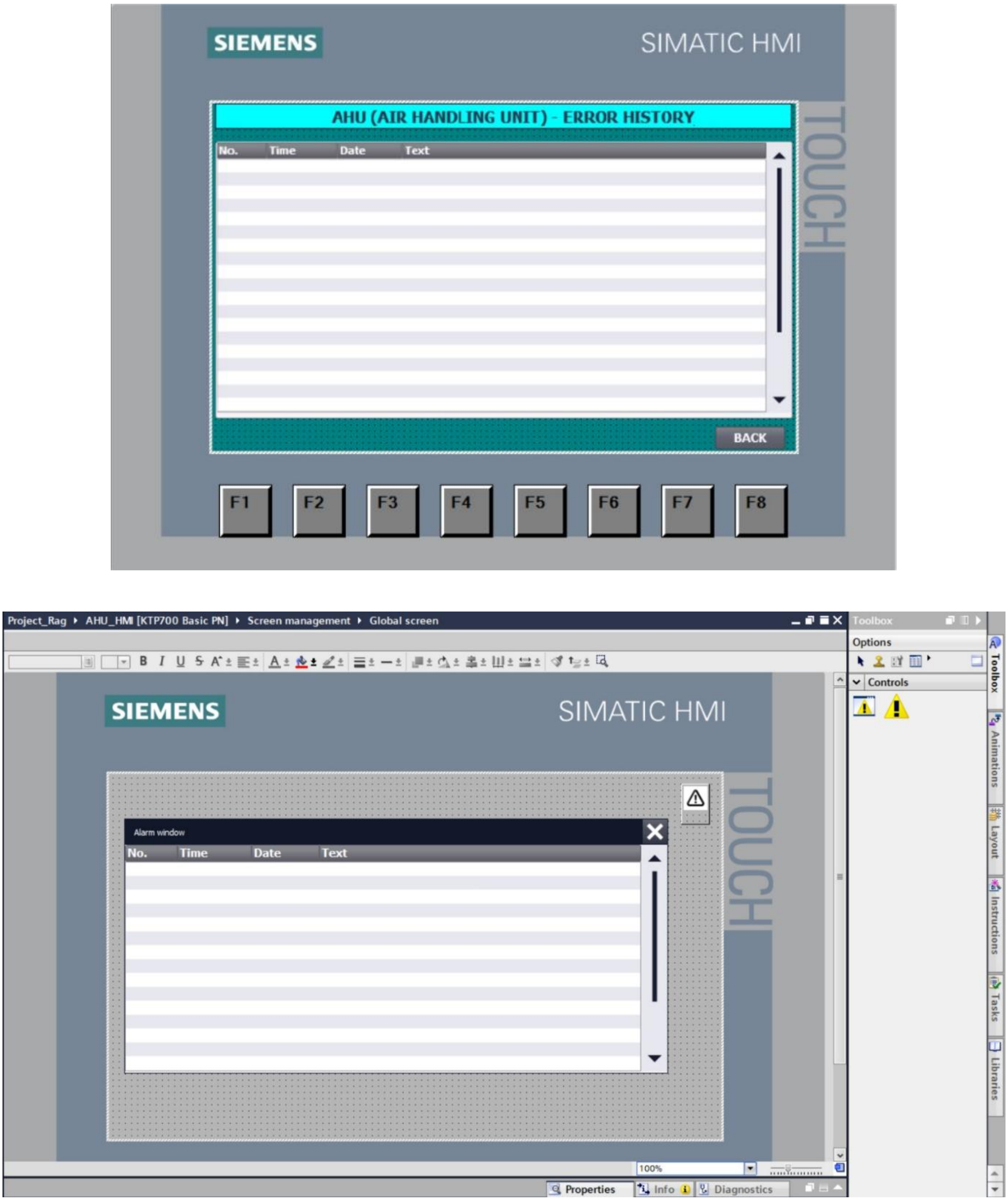
**DEVELOPMENT OF HUMAN MACHINE INTERFACE**

To simplify operator interaction with the Air Handling Unit, a dedicated Human Machine Interface was developed and integrated with the PLC. The HMI provides a graphical representation of the complete control system and allows the operator to perform all necessary operations through a single interface. The home screen of the HMI contains Start and Stop controls along with the available speed selection options. Separate indicators are provided to display the current operating status of the blower, selected speed level, communication status, and system availability.

Additional indication lamps were incorporated to display important operational information such as system running condition, blower status, emergency stop activation, communication health, and fault indication. The graphical layout was designed to be simple and intuitive so that operators could understand the operating condition of the system without requiring extensive technical knowledge. Another important advantage of the HMI is that maintenance personnel can quickly identify operating conditions during troubleshooting without opening the electrical control panel.



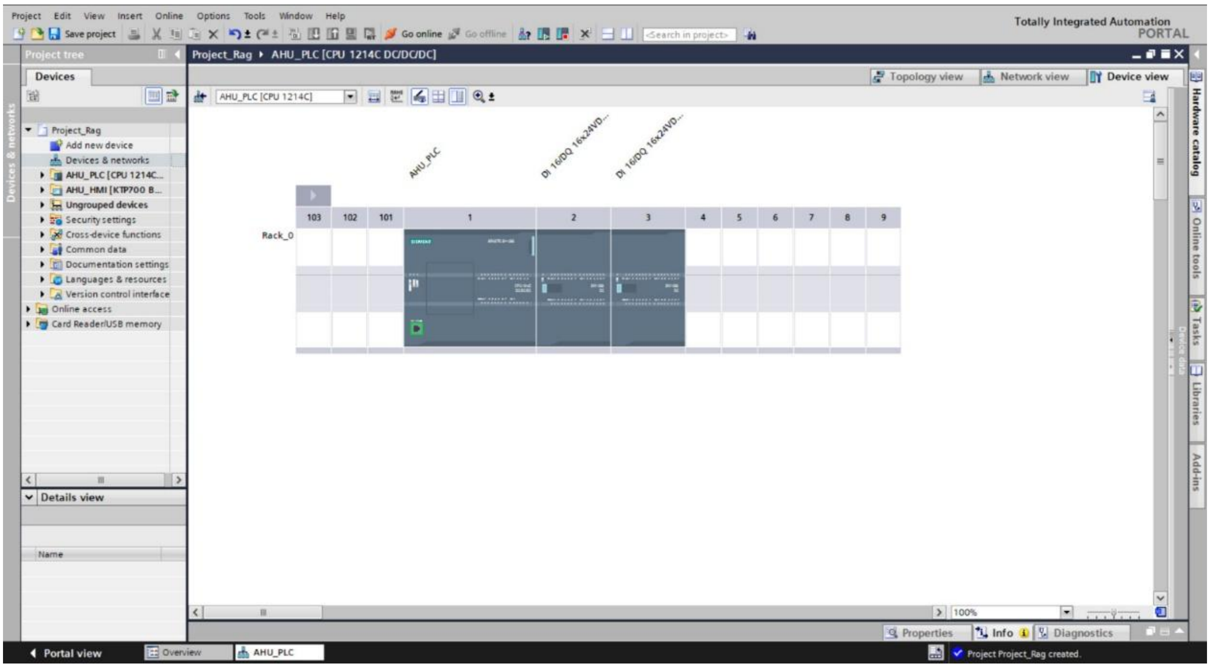



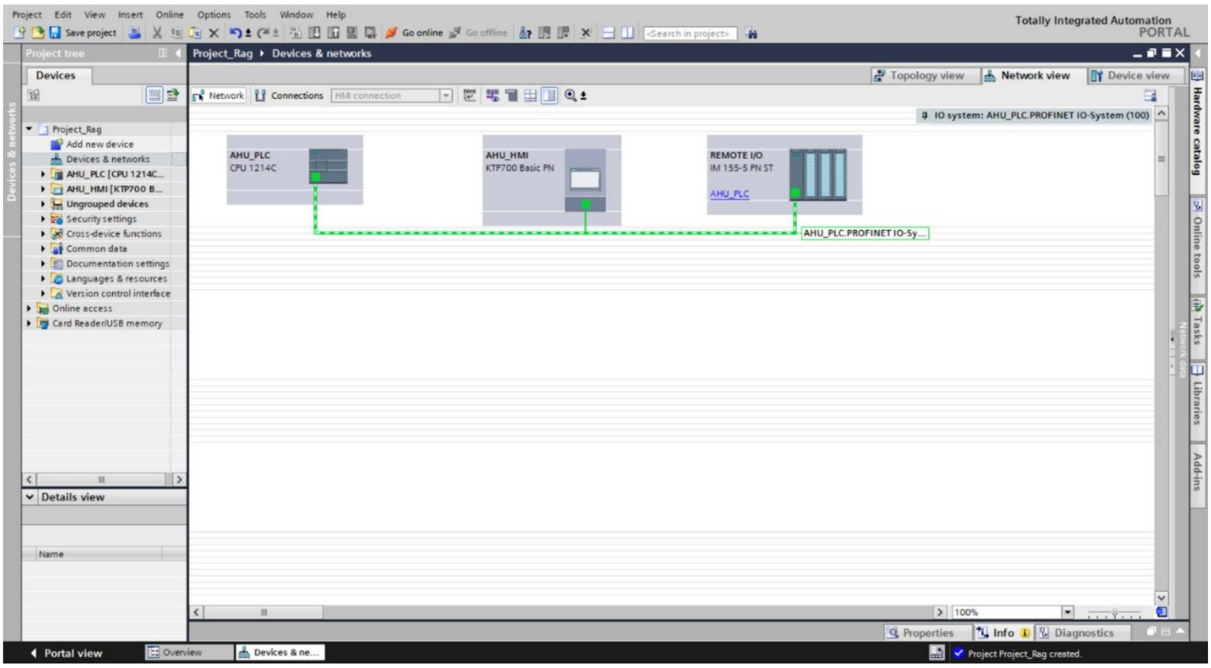



*Figures of the HMI Design and Development*

## COMMUNICATION BETWEEN PLC AND HMI

Reliable communication between the PLC and the Human Machine Interface is essential for ensuring proper operation of the automation system. The HMI continuously exchanges data with the PLC by reading internal memory locations, input signals, output status, and process variables. Whenever the operator presses a button on the HMI, the command is transmitted to the PLC through the communication network. The PLC processes the command according to the programmed logic and updates the corresponding outputs. Simultaneously, the PLC sends updated status information back to the HMI, allowing the operator to observe the real-time operating condition of the Air Handling Unit. This continuous exchange of information ensures synchronization between operator commands and machine operation while improving overall usability of the control system.




*Figures of the PLC-HMI Communication Configuration*

## TESTING AND VERIFICATION OF THE CONTROL LOGIC

After completion of the PLC program and HMI development, comprehensive testing was carried out before implementation of the control panel. Initially, offline simulation was performed within the TIA Portal software to verify the correctness of the ladder logic. Individual inputs were activated sequentially to ensure that the corresponding outputs responded according to the intended control sequence. Following successful software verification, hardware testing was performed by energizing the PLC and verifying every input and output connection individually. Push buttons, selector switches, relays, indication lamps, and communication modules were tested to confirm proper operation.

Finally, the manual speed selection sequence was tested under different operating conditions. Each speed level was verified individually to ensure correct output activation, while interlocking conditions were also checked to confirm that only one speed remained active at a time. The Emergency Stop circuit was tested separately to verify immediate shutdown of all outputs under emergency conditions. Successful completion of these tests confirmed that the developed PLC control system satisfied the functional requirements established during the project planning stage.

## CHALLENGES ENCOUNTERED DURING THE PROJECT

The development of the PLC-based Air Handling Unit control panel involved several engineering challenges that required systematic analysis and problem-solving throughout the project. Although the overall control philosophy was based on manual speed selection, considerable effort was required to ensure that the system operated safely, reliably, and in accordance with industrial standards. One of the primary challenges was understanding the existing control system before developing the new PLC-based solution. A detailed study of the existing electrical wiring, control sequence, and operational requirements was necessary to ensure that the proposed system could perform all the required functions without affecting the normal operation of the Air Handling Unit.

Another important challenge involved selecting suitable PLC input and output addresses while ensuring proper mapping between field devices and the controller. Every signal had to be carefully assigned and verified to prevent addressing conflicts during program execution. The development of the ladder logic also required careful planning to ensure that only one speed selection remained active at any given time. Proper interlocking conditions had to be incorporated within the program to prevent simultaneous activation of multiple outputs, thereby protecting the blower motor and associated electrical equipment. The preparation of the electrical wiring demanded a high level of accuracy. Every terminal connection had to be checked carefully against the electrical schematic to eliminate wiring errors before power was applied to the control panel. Proper wire identification, cable routing, ferrule numbering, and terminal labelling were maintained throughout the panel to improve future maintenance activities. Testing the complete system required repeated verification of each input, output, relay operation, communication signal, and

HMI screen. Minor programming modifications and signal verification were carried out during the testing phase to ensure smooth operation of the complete automation system. Despite these challenges, systematic engineering practices, guidance from maintenance engineers, and continuous testing enabled the successful completion of the project.

## RESULTS AND ACHIEVEMENTS

The successful development of the PLC-based Air Handling Unit control panel demonstrated the practical application of industrial automation techniques within a manufacturing environment. The completed system provided a reliable and organized method of controlling the Air Handling Unit while simplifying operation and maintenance. The PLC-based control system replaced conventional control methods with programmable automation, thereby improving operational flexibility and making future modifications considerably easier. The implementation of manual speed selection enabled the operator to select the required blower speed through the Human Machine Interface while allowing the PLC to execute the corresponding control sequence accurately.

The Human Machine Interface improved operator convenience by providing a centralized interface for machine operation and status monitoring. Real-time indications of system conditions enhanced operator awareness and reduced the time required for fault identification. The organized electrical panel layout, systematic wiring practices, and proper component identification improved maintenance accessibility and simplified future troubleshooting activities. The modular nature of the PLC system also allows additional automation features to be incorporated in future without significant hardware modifications. The project demonstrated that PLC-based automation offers significant advantages in terms of reliability, maintainability, flexibility, and operational efficiency when compared with conventional relay-based control systems.

## ADVANTAGES OF THE DEVELOPED PLC-BASED AHU CONTROL PANEL

The implementation of the PLC-based Air Handling Unit control panel offers several technical and operational advantages over conventional control methods. Some of the major advantages observed are listed below:

- Simplified operation through a user-friendly Human Machine Interface.
- Reliable manual speed selection with proper interlocking logic.
- Reduced complexity of electrical control wiring.
- Easier troubleshooting using PLC diagnostics and online monitoring.
- Improved operational reliability due to programmable control.
- Enhanced flexibility for future software modifications without major hardware changes.
- Better organization of electrical components inside the control panel.

- Improved maintenance accessibility through proper labelling and documentation.
- Faster fault identification using PLC status monitoring and HMI indications.
- Modular architecture allowing future expansion of the automation system.
- Improved safety through programmed interlocks and emergency stop functionality.
- Reduced maintenance effort compared with conventional relay-based control circuits.
- Better documentation of control logic, wiring diagrams, and system configuration.
- Increased operational efficiency and ease of system monitoring.

## FUTURE SCOPE

Although the developed system satisfies the present operational requirements of the Air Handling Unit, several advanced automation features can be incorporated in future to further improve system performance and energy efficiency. The manual speed selection implemented during the project can be upgraded to automatic speed control by integrating temperature sensors, humidity sensors, differential pressure sensors, or air quality sensors. The PLC can then automatically regulate blower speed according to the environmental conditions within the shop floor.

Variable Frequency Drives (VFDs) can also be incorporated to achieve smooth and energy-efficient motor speed control. This would reduce electrical power consumption while improving the overall efficiency of the ventilation system. Integration of the PLC with Supervisory Control and Data Acquisition (SCADA) systems would enable centralized monitoring, historical data logging, alarm management, and remote diagnostics from the maintenance control room.

Industrial Ethernet communication can further be utilized for integrating multiple Air Handling Units into a centralized building management system, thereby improving overall plant monitoring and control. The incorporation of predictive maintenance techniques based on vibration monitoring, motor current analysis, bearing temperature monitoring, and equipment health diagnostics can also improve machine reliability while reducing unexpected failures. These future developments demonstrate the scalability of the PLC-based automation system developed during the project.

## TECHNICAL KNOWLEDGE AND SKILLS ACQUIRED

The Air Handling Unit automation project provided valuable practical exposure to industrial automation engineering and significantly enhanced technical competency. The major knowledge and skills acquired during the project are summarized below:

- Understanding the construction and working principle of industrial Air Handling Units.
- Studying industrial HVAC applications within manufacturing environments.
- Understanding the complete engineering workflow involved in automation projects.

- Practical exposure to Siemens PLC hardware and industrial automation architecture.
- Development of PLC programs using Siemens Totally Integrated Automation (TIA) Portal.
- Understanding PLC hardware configuration and input-output addressing.
- Preparation of electrical wiring based on industrial electrical schematics.
- Selection and integration of industrial electrical control components.
- Development of Human Machine Interface (HMI) screens for operator interaction.
- Understanding PLC-HMI communication and data exchange.
- Implementation of manual speed selection control logic.
- Testing, debugging, and verification of industrial PLC programs.
- Understanding electrical panel design and component arrangement.
- Improvement of troubleshooting and logical problem-solving skills.
- Enhancement of documentation, engineering analysis, and systematic project execution skills.
- Practical understanding of industrial automation standards and engineering practices.

# PROJECT WORK 2 – PLC-BASED COOLANT CONDITIONING SYSTEM FOR SHOP FLOOR 5

## INTRODUCTION

As part of the industrial internship in the Maintenance Department at Ashok Leyland Limited, an opportunity was provided to participate in the development of a PLC-based automation system for the Coolant Conditioning System installed in Shop Floor 5. The project focused on developing an industrial automation solution capable of monitoring important operating parameters while providing reliable control of the coolant conditioning process. The coolant conditioning system plays a vital role in maintaining the quality and operating condition of coolant used during engine testing. During continuous engine operation, coolant absorbs heat generated by the engine and helps maintain its operating temperature within the permissible range. Repeated usage of coolant results in changes in temperature, contamination levels, and overall cooling efficiency. Therefore, the coolant must be continuously circulated, monitored, and conditioned before being supplied again for engine testing operations.

The project involved studying the existing coolant conditioning process, understanding the operational sequence of the system, analysing instrumentation requirements, developing the PLC control program, configuring the Human Machine Interface (HMI), integrating field devices, monitoring rotational speed parameters, and verifying overall system operation. Unlike routine maintenance activities, this project involved the complete engineering lifecycle of an industrial automation system, including requirement analysis, hardware selection, software development, instrumentation integration, testing, and commissioning. The project provided practical exposure to industrial automation techniques employed in manufacturing industries while strengthening knowledge of PLC programming, industrial instrumentation, electrical control systems, and process automation.

## BACKGROUND OF THE PROJECT

Modern engine testing facilities require coolant to maintain controlled operating temperatures throughout engine performance evaluation. Continuous circulation of properly conditioned coolant prevents overheating of the engine while ensuring repeatable and accurate testing conditions. The existing coolant conditioning arrangement was capable of supplying coolant to the engine test beds with U90 PLC; however, there was a need to improve monitoring, automation, and operator interaction by incorporating PLC-based control. Accurate monitoring of process parameters such as pump operation, motor speed, coolant circulation, and equipment status would improve operational reliability while simplifying maintenance activities. The project therefore focused on developing a centralized PLC-based control system capable of monitoring equipment operation, displaying important process parameters, and improving the maintainability of the coolant conditioning system.

## NEED FOR THE PROJECT

The coolant conditioning system operates continuously during engine testing and therefore requires reliable monitoring and control. Conventional control methods provide limited diagnostic information and often make troubleshooting more difficult during equipment failures. The implementation of a PLC-based automation system offers several advantages, including improved process monitoring, better visualization through the Human Machine Interface, simplified troubleshooting, accurate parameter calculation, reliable equipment control, and easy future expansion of the automation system.

Another important requirement was the continuous monitoring of rotating equipment associated with the coolant conditioning process. Accurate determination of rotational speed allows maintenance engineers to verify equipment performance and quickly identify abnormal operating conditions before major failures occur. The project therefore aimed to improve operational reliability while introducing modern industrial automation practices into the coolant conditioning process.

## OBJECTIVES OF THE PROJECT

The major objectives of the Coolant Conditioning System project are summarized below.

- To study the working principle of the coolant conditioning system.
- To understand the coolant circulation process during engine testing.
- To analyse the existing control methodology.
- To develop a PLC-based monitoring and control system.
- To configure Siemens PLC hardware.
- To develop the PLC program using TIA Portal.
- To configure the Human Machine Interface.
- To monitor Engine RPM and Motor RPM.
- To implement High-Speed Counter technology.
- To perform accurate RPM calculation.
- To improve operator monitoring.
- To simplify maintenance and troubleshooting.
- To verify complete system operation through testing.

## WORKING PRINCIPLE OF THE COOLANT CONDITIONING SYSTEM

The coolant conditioning system is designed to circulate coolant continuously between the storage tank, circulation pumps, conditioning equipment, and engine test beds. During engine testing, coolant absorbs heat generated by the engine and carries this thermal energy away from the engine block. The heated coolant is then returned to the conditioning system where it is prepared for reuse. The coolant first enters the storage or collection section before being circulated by electrically driven centrifugal pumps. Depending upon the system configuration, the coolant may pass through filtration units, heat exchangers, or cooling equipment where excess heat is removed and coolant quality is maintained.

Once the coolant reaches the desired operating condition, it is pumped back towards the engine test beds for the next testing cycle. Throughout this process, continuous circulation ensures stable engine operating temperature while preventing overheating during prolonged engine testing. The developed PLC system continuously monitors important equipment associated with this circulation process. The operating status of pumps, motors, and rotating equipment is monitored in real time, enabling maintenance personnel to observe the health of the complete coolant conditioning system.

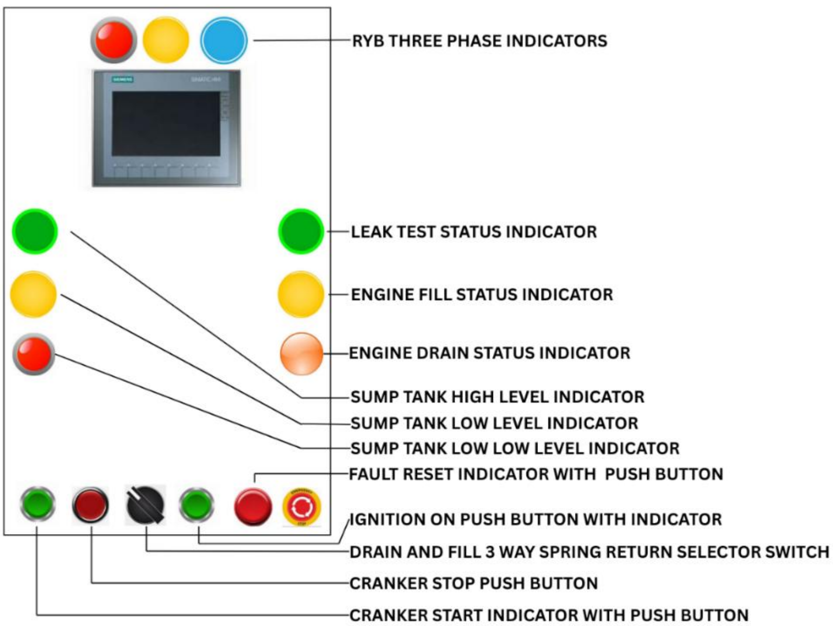



*Figure of the PLC Panel Front Buttons and Indication System*

## OVERALL CONTROL PHILOSOPHY

The overall control philosophy adopted for this project was based on reliable monitoring, accurate parameter calculation, and simplified operator interaction. The PLC continuously acquires signals from various field devices connected to the coolant conditioning system. These signals are processed using the programmed control logic to determine equipment operating conditions and calculate important engineering parameters.

One of the major features incorporated within the project is the continuous monitoring of Engine RPM and Motor RPM. High-speed pulse signals generated by the rotating equipment are processed using dedicated High-Speed Counter technology available within the Siemens PLC. The calculated RPM values are displayed on the Human Machine Interface, enabling operators and maintenance engineers to continuously monitor equipment performance. The PLC also updates the HMI with real-time operating information, allowing quick identification of abnormal conditions and improving maintenance efficiency. The complete system was designed with future expansion capability, enabling additional sensors and monitoring functions to be incorporated whenever required.

## MAJOR COMPONENTS OF THE COOLANT CONDITIONING SYSTEM

The Coolant Conditioning System consists of several mechanical, electrical, and automation components that work together to ensure the continuous circulation and conditioning of coolant used during engine testing. Every component performs a specific function that contributes to maintaining proper coolant flow, stable operating temperature, and reliable system performance. The major components of the system include coolant storage tanks, centrifugal circulation pumps, electric motors, piping network, isolation valves, pressure gauges, temperature sensors, flow monitoring devices, electrical control panels, Siemens Programmable Logic Controller (PLC), Human Machine Interface (HMI), communication network, and various electrical protection devices. Each component was studied individually to understand its operating principle, function within the overall system, and interaction with the automation system. Understanding the role of every component was essential before developing the PLC program and integrating the control system.

## COOLANT CIRCULATION PUMP

The circulation pump is one of the most important components of the coolant conditioning system. Its primary function is to continuously circulate coolant between the storage tank, conditioning unit, and engine test beds during engine testing operations. The pump draws coolant from the storage section and supplies it through the piping network to the engine under controlled flow conditions. After absorbing heat from the engine, the coolant returns to the conditioning system for recirculation.

Continuous pump operation is essential because any interruption in coolant flow may result in overheating of the engine, affecting both equipment safety and testing accuracy. The pump is driven by an electric motor whose operating condition is continuously monitored by the automation system. Proper monitoring of pump performance enables maintenance personnel to identify abnormal operating conditions before they develop into major failures.

## ELECTRIC MOTOR AND DRIVE SYSTEM

The circulation pump is mechanically coupled to an electric motor that provides the necessary mechanical power required for coolant circulation. During operation, the motor rotates the pump impeller, thereby generating the pressure required to circulate coolant throughout the system. The operating condition of the motor directly influences coolant flow rate and overall system performance. Therefore, continuous monitoring of motor operation is an important requirement for maintaining reliable coolant circulation.

The automation system developed during the project continuously monitors the motor operating condition and rotational speed. Accurate monitoring of motor speed enables maintenance engineers to verify whether the motor is operating under normal conditions and assists in identifying mechanical or electrical abnormalities. Industrial motors operating under continuous duty require proper monitoring to prevent excessive wear, overheating, vibration, and unexpected failures. Integrating motor monitoring within the PLC system therefore improves equipment reliability while simplifying maintenance activities.

## INDUSTRIAL INSTRUMENTATION USED

Industrial instrumentation provides the necessary process information required by the PLC to monitor and control the coolant conditioning process. Various sensors installed throughout the system continuously transmit information regarding operating conditions, allowing the PLC to process the data and provide real-time visualization through the Human Machine Interface. Temperature sensors monitor coolant temperature at different stages of the circulation process. Maintaining the coolant within the desired temperature range is essential for ensuring consistent engine testing conditions.

Pressure measuring devices monitor coolant pressure within the piping network and help verify proper pump operation. Abnormal pressure values may indicate pump malfunction, pipeline blockage, leakage, or valve-related problems. Flow monitoring devices ensure that coolant circulation is maintained continuously during operation. Adequate coolant flow is essential for efficient heat removal from the engine.

In addition to process sensors, rotational speed sensors are installed on the engine and motor to generate pulse signals corresponding to shaft rotation. These high-speed pulse signals are processed by the PLC for calculating Engine RPM and Motor RPM. The integration of industrial instrumentation significantly improves process visibility, equipment monitoring, and maintenance efficiency.

| PARAMETER | SPECIFICATION OF PRESSURE SENSOR |
|---|---|
| Sensor Model | UNIK 5000 |
| Measurement Range | 0 to 2.5 bar |
| Supply Voltage | 7 V DC to 32 V DC |
| Output Signal | 4–20 mA |
| Operating Temperature | -20°C to 80°C |

**SIEMENS PLC HARDWARE**

The Siemens Programmable Logic Controller forms the central control unit of the coolant conditioning system. All field devices communicate with the PLC, which processes the incoming signals according to the programmed logic and performs the necessary calculations before generating output signals and updating the Human Machine Interface. The PLC continuously scans all digital and analogue inputs, executes the control program, updates output signals, and performs communication with the HMI. This cyclic execution enables real-time monitoring of equipment operating conditions.

The modular architecture of the Siemens PLC provides flexibility for incorporating additional input and output modules whenever future expansion becomes necessary. This feature ensures that the automation system can be upgraded without replacing the existing controller. Another important feature utilized during this project is the High-Speed Counter (HSC) capability available within the PLC. Unlike ordinary digital inputs, High-Speed Counters are capable of accurately processing rapidly changing pulse signals generated by rotating equipment. This capability forms the basis for accurate Engine RPM and Motor RPM measurement.

**HUMAN MACHINE INTERFACE (HMI)**

The Human Machine Interface provides a graphical platform through which operators and maintenance engineers interact with the coolant conditioning system. The HMI continuously displays important operating parameters such as equipment running status, pump condition, Engine RPM, Motor RPM, communication status, alarm indications, and other system information received from the PLC.

Operators can easily observe system behaviour without opening the electrical control panel. During maintenance activities, the HMI also assists engineers in identifying abnormal operating conditions by displaying real-time parameter values. The graphical interface was designed to present information in a clear and organized manner, thereby improving operator convenience while reducing the time required for troubleshooting.

**ELECTRICAL CONTROL PANEL**

The electrical control panel houses all automation and power control components required for operating the coolant conditioning system. Proper arrangement of electrical equipment inside the panel is essential for ensuring safe operation, reliable performance, and simplified maintenance. The panel consists of miniature circuit breakers, switched-mode power supplies, PLC modules, relays, contactors, terminal blocks, communication cables, indication lamps, and protective devices. Every component is mounted systematically on DIN rails with adequate spacing for heat dissipation and future maintenance.

Cable ducts are provided for routing all wiring in an organized manner. Each conductor is properly numbered using ferrules, while terminal blocks are clearly identified according to the electrical schematic. The front panel accommodates the Human Machine Interface and indication lamps, allowing operators to monitor system status without opening the enclosure. Proper panel design improves accessibility, minimizes wiring errors, simplifies troubleshooting, and enhances the overall reliability of the automation system.

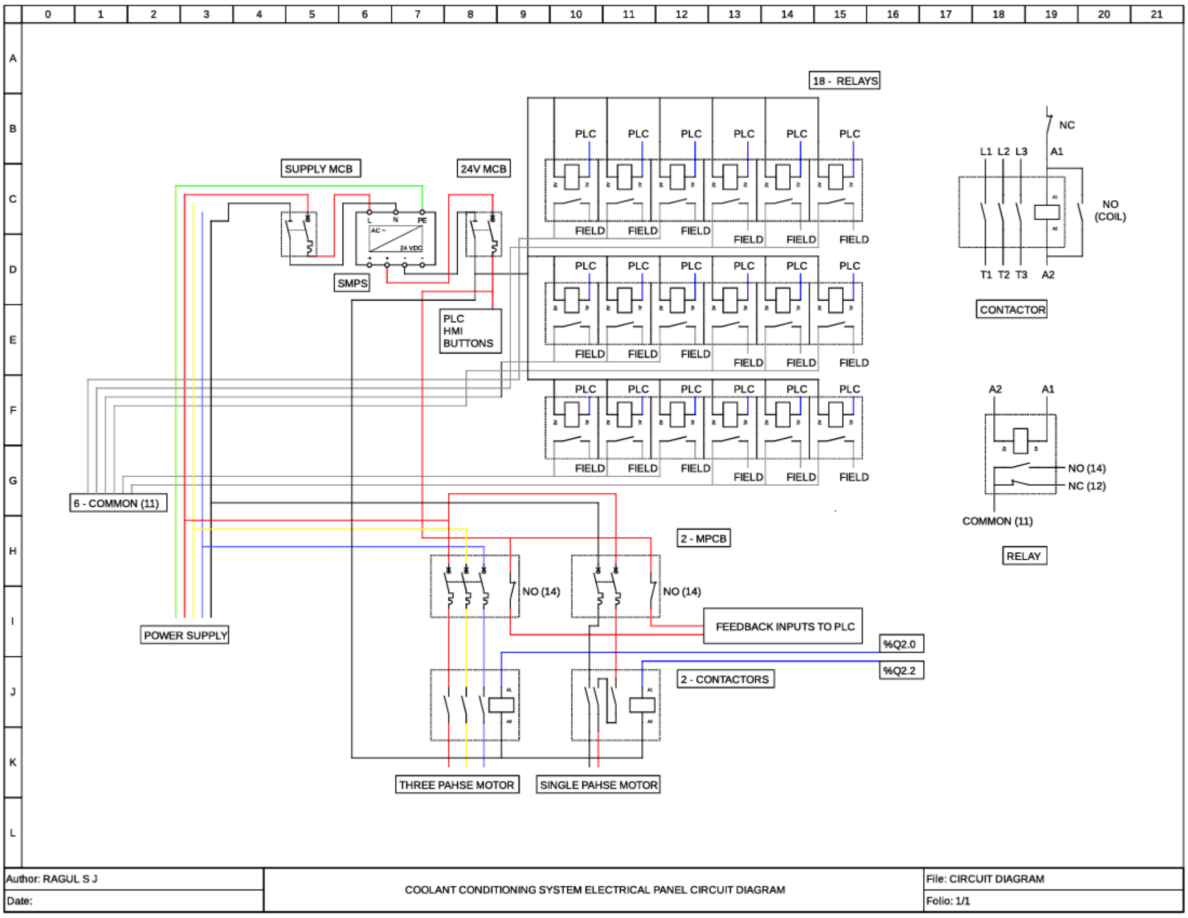
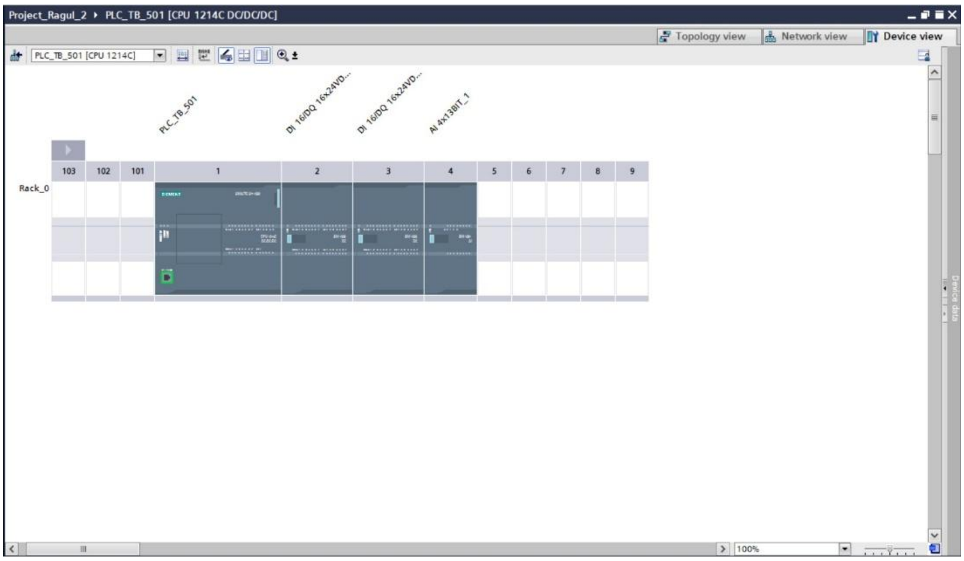


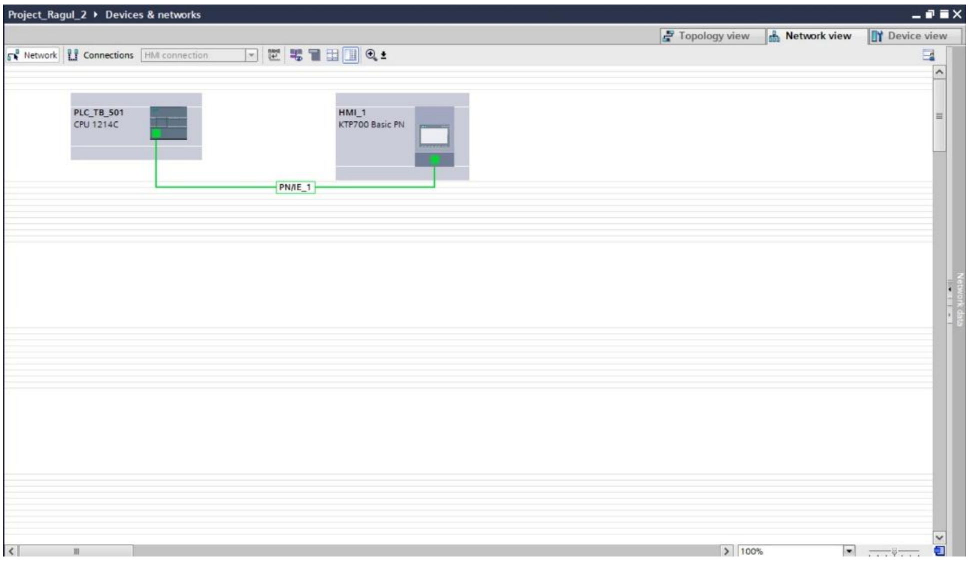

*Figure of the Panel Circuit Diagram*

## SYSTEM ARCHITECTURE AND SIGNAL FLOW

The complete automation system follows a structured signal flow beginning from the field devices and ending at the operator interface. Initially, field sensors installed on the engine, motor, and coolant conditioning equipment generate electrical signals corresponding to their operating conditions. These signals are transmitted to the PLC through appropriate input modules. The PLC continuously processes these signals using the developed control program. High-speed pulse signals received from the rotational speed sensors are processed using dedicated High-Speed Counter hardware, while other process signals are evaluated according to the programmed control logic.

After processing the acquired information, the PLC updates the Human Machine Interface with real-time operating data. Operators can therefore continuously monitor the operating condition of the coolant conditioning system while maintenance engineers can use the displayed information for troubleshooting and performance evaluation. This systematic signal flow ensures reliable acquisition, processing, visualization, and monitoring of important process parameters throughout the coolant conditioning system.




*Figures of the PLC-HMI Communication Configuration*

## DEVELOPMENT OF THE PLC PROGRAM

The development of the PLC program formed the core of the Coolant Conditioning System project. After understanding the working principle of the complete system and identifying the required field devices, the next phase involved developing the control logic using Siemens Totally Integrated Automation (TIA) Portal. The objective of the program was to continuously monitor important operating parameters of the coolant conditioning system while accurately calculating the rotational speed of both the engine and the coolant circulation motor.

The PLC program was developed in a structured and modular manner to simplify troubleshooting, future modifications, and maintenance. Individual sections of the program were created for system initialization, communication, high-speed pulse acquisition, RPM calculation, Human Machine Interface data transfer, alarm monitoring, and equipment status indication.

Unlike conventional PLC applications that primarily use digital inputs and outputs for ON/OFF control, this project required the PLC to process high-frequency pulse signals generated by rotating equipment. Since these pulse signals change at a very high rate, ordinary PLC scan cycles cannot guarantee accurate counting. Therefore, High-Speed Counter (HSC) technology available within the Siemens PLC was utilized for acquiring the pulse signals without any loss of information. The complete PLC program was verified through simulation and online testing before being integrated with the coolant conditioning system.

## HIGH-SPEED COUNTER (HSC) IMPLEMENTATION

One of the most important technical aspects of this project was the implementation of High-Speed Counter (HSC) technology for measuring the rotational speed of the engine and coolant circulation motor. In industrial automation systems, rotating machines generally produce pulse signals through speed sensors or encoders mounted on the rotating shaft. Every rotation generates a specific number of electrical pulses depending upon the Pulse Per Revolution (PPR) of the sensing device. These pulses can then be processed by the PLC to determine the actual rotational speed of the machine.

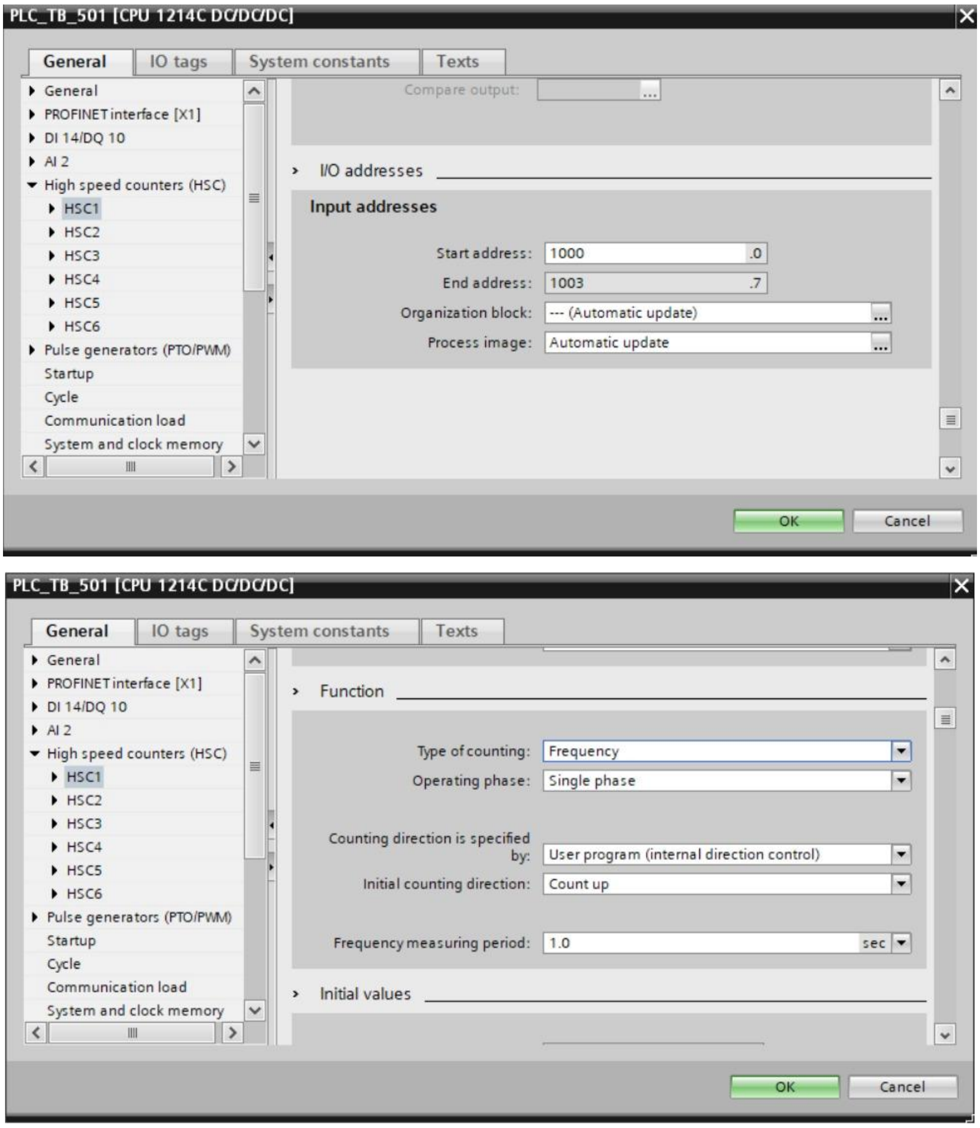
However, the pulse frequency generated during machine operation can become extremely high, particularly at higher rotational speeds. Under such conditions, ordinary digital input scanning performed by the PLC is insufficient because the PLC scan time may be longer than the pulse duration. This results in missed pulses and inaccurate RPM calculations. To overcome this limitation, Siemens PLCs provide dedicated High-Speed Counter hardware capable of independently counting pulses irrespective of the normal PLC scan cycle. The High-Speed Counter continuously counts every incoming pulse at the hardware level and provides highly accurate counting even at very high rotational speeds.

For this project, two separate High-Speed Counter channels were utilized.

**ID1000** was configured for monitoring the Engine RPM.

**ID1004** was configured for monitoring the Motor RPM.

Both High-Speed Counter channels continuously counted the incoming pulses from their respective sensors and supplied the accumulated count values to the PLC program for further processing. The implementation of dedicated High-Speed Counters significantly improved measurement accuracy and ensured reliable monitoring of both rotating systems under all operating conditions.




*Figures of the Configuration of HSC*

## CYCLIC INTERRUPT-BASED DATA ACQUISITION

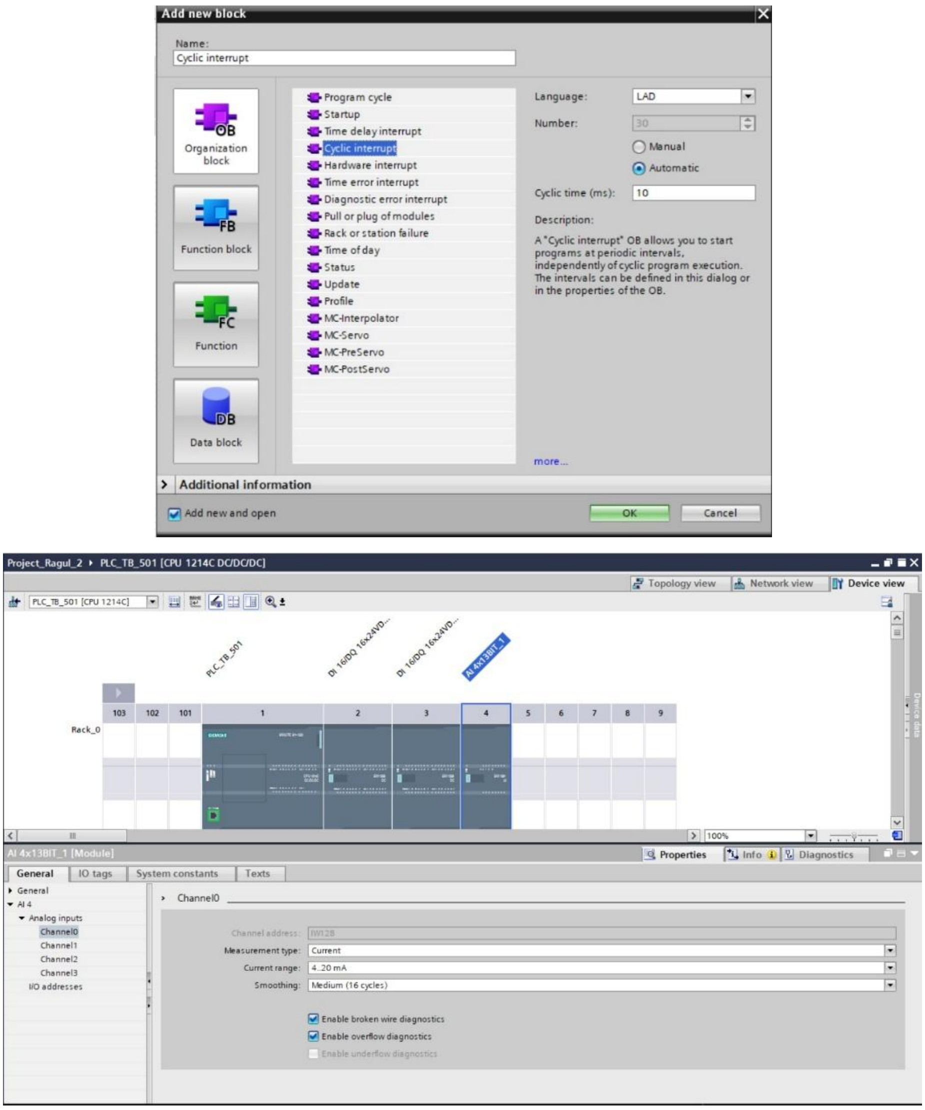
To achieve accurate RPM measurement, the pulse count obtained from the High-Speed Counter must be read at precisely defined intervals. Instead of performing the calculations within the normal cyclic program, a Cyclic Interrupt Organization Block was employed. The Cyclic

Interrupt executes the RPM calculation routine at a fixed execution interval of **less than 10 milliseconds**. This deterministic execution ensures that pulse data is acquired at regular intervals irrespective of the PLC scan time or execution of other program sections.

Using a Cyclic Interrupt offers several advantages:

- Uniform sampling interval for pulse acquisition.
- Improved measurement accuracy.
- Faster response to speed variations.
- Reduced calculation errors.
- Stable real-time monitoring of rotating equipment.

Every execution of the Cyclic Interrupt reads the latest High-Speed Counter values, calculates the pulse frequency, converts the frequency into rotational speed, and updates the calculated values for display on the Human Machine Interface.




*Figures of the Configuration of Cyclic Interrupt Block and Analog Inputs*

## SELECTION OF DATA TYPES

Since the pulse count obtained from the High-Speed Counter increases rapidly during machine operation, careful selection of PLC data types was essential. The **Double Integer (DINT)** data type was selected for storing pulse counts, intermediate calculations, and RPM values. Compared with standard Integer variables, the Double Integer provides a much larger numerical range and therefore prevents overflow during high-speed counting operations. The use of DINT variables ensured stable program execution even when the rotating equipment operated at higher speeds for extended periods. Proper selection of data types also improved calculation accuracy and enhanced overall reliability of the PLC program.

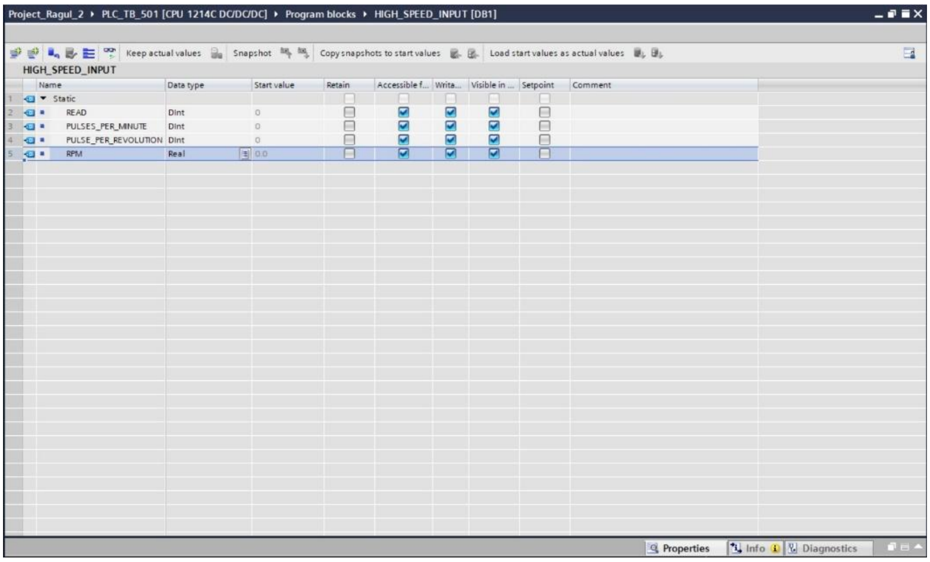



*Figure of the Datatype Selection in Data Block*

## RPM CALCULATION METHODOLOGY

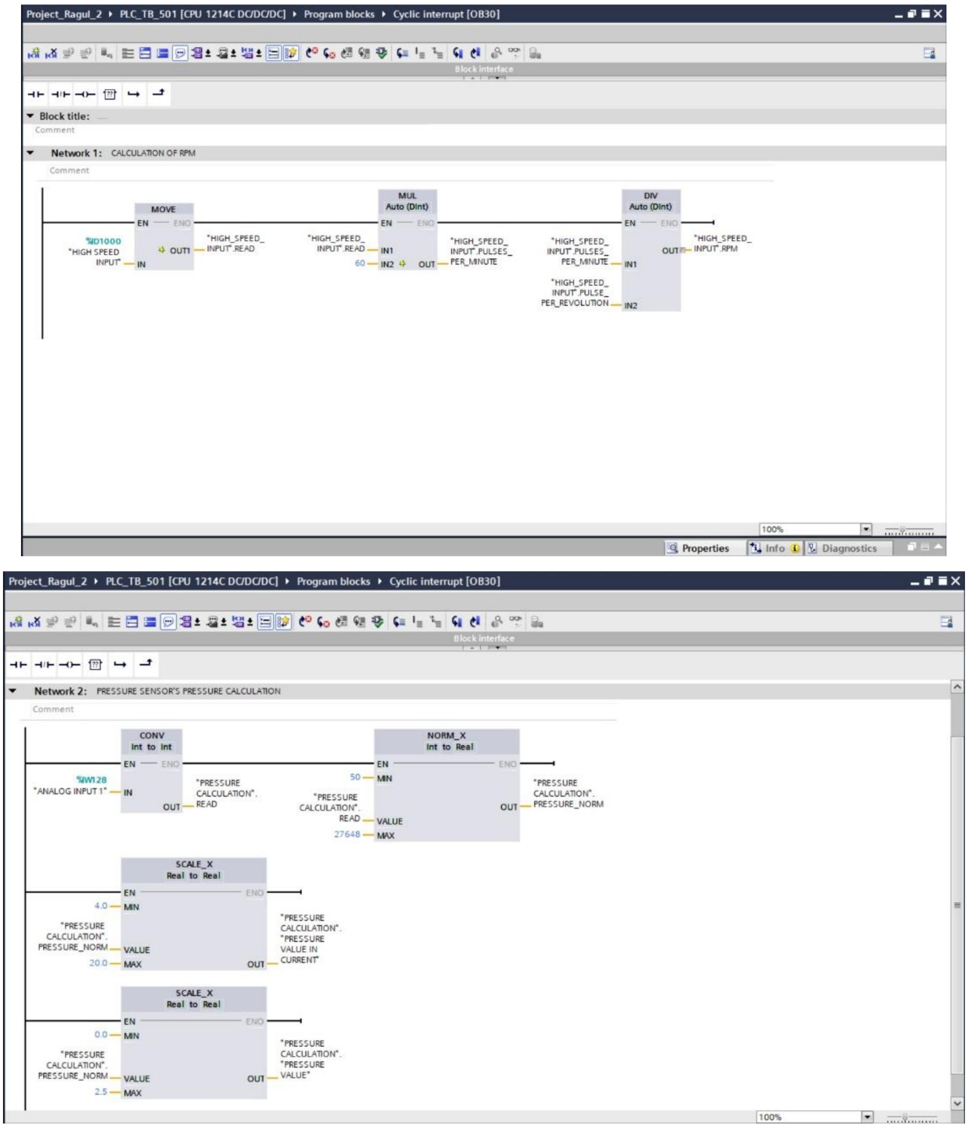
The primary purpose of the High-Speed Counter implementation was to determine the rotational speed of both the engine and the coolant circulation motor in terms of Revolutions Per Minute (RPM). Initially, the High-Speed Counter continuously accumulates the incoming pulse count generated by the speed sensor. During every execution of the Cyclic Interrupt, the pulse count recorded during the sampling interval is used to determine the pulse frequency. The frequency represents the number of pulses generated per second and forms the basis for RPM calculation. Once the pulse frequency is obtained, the PLC calculates the rotational speed using the following relationship:

$$\mathbf{RPM = (Frequency * 60) / PPR}$$

where:

- Frequency = Number of pulses received per second
- 60 = Conversion from seconds to minutes
- Pulse Per Revolution (PPR) = Number of pulses for one complete shaft revolution

The same calculation procedure was independently performed for both the Engine RPM and Motor RPM. The calculated RPM values were continuously updated within the PLC memory and transmitted to the Human Machine Interface for real-time monitoring.



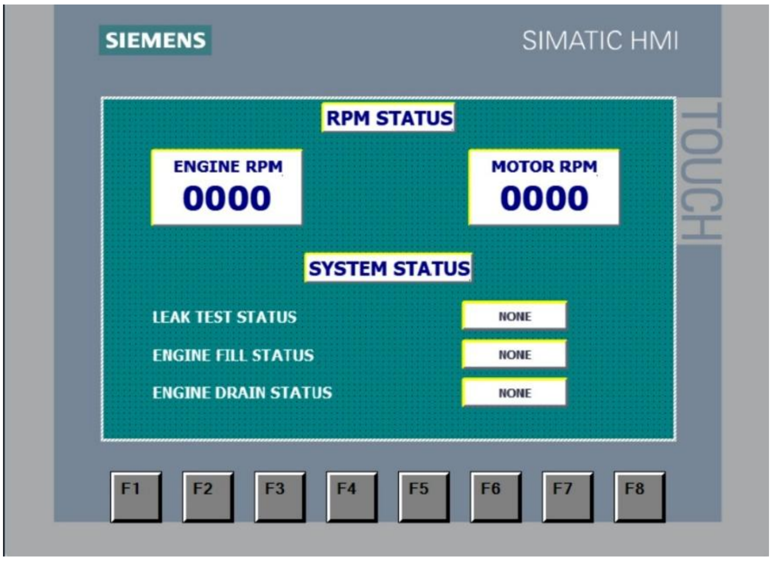
*Figures of the Logic of RPM Calculation and Pressure Calculation*

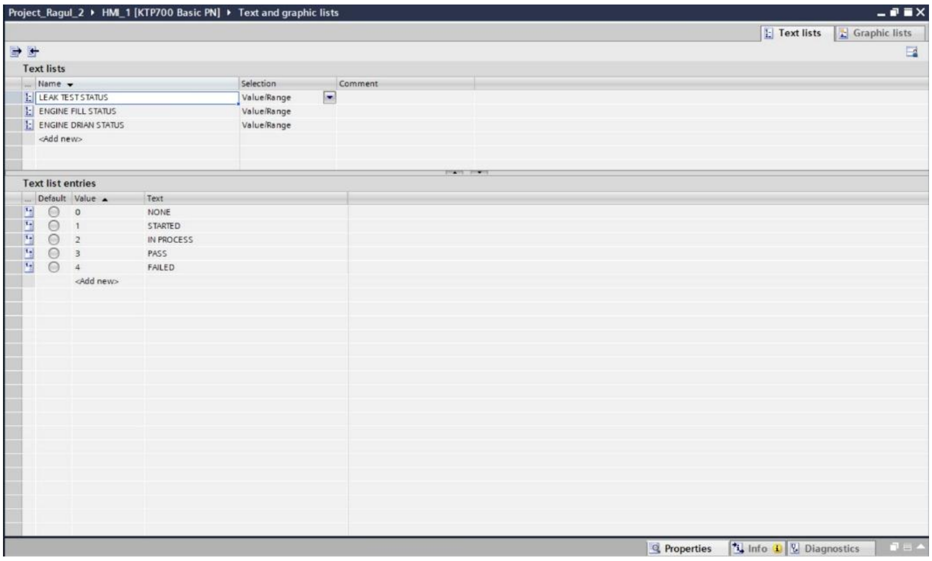
## ENGINE RPM MONITORING

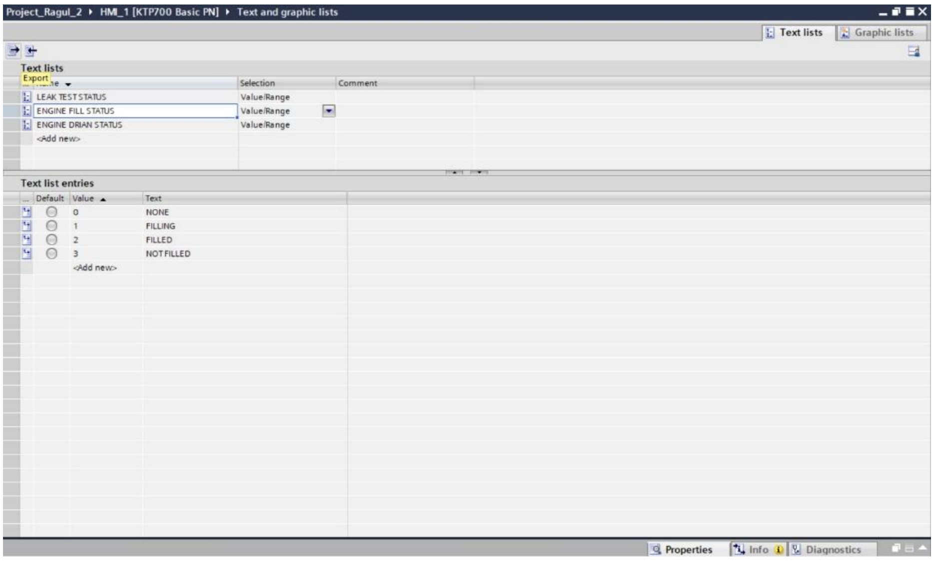
Continuous monitoring of Engine RPM forms one of the most important functions of the developed automation system. Accurate knowledge of engine speed enables maintenance engineers to verify proper engine operation during coolant conditioning and engine testing processes. The Engine RPM signal was obtained through the High-Speed Counter configured as **ID1000**. The incoming pulse signal was continuously counted by the dedicated hardware counter, eliminating the possibility of missed pulses. The PLC periodically read the accumulated pulse count through the Cyclic Interrupt routine and converted it into RPM using the programmed calculation algorithm. The calculated Engine RPM was continuously transmitted to the Human Machine Interface where it was displayed in real time. This enabled operators to observe engine operating conditions continuously throughout the testing process.

## MOTOR RPM MONITORING

In addition to Engine RPM, the rotational speed of the coolant circulation motor was also monitored continuously. A separate High-Speed Counter configured as **ID1004** acquired the pulse signals generated by the motor speed sensor. Similar to the Engine RPM calculation, the pulse frequency was determined periodically within the Cyclic Interrupt routine. The Motor RPM was then calculated using the programmed mathematical relationship and stored within PLC memory before being transmitted to the Human Machine Interface. Real-time monitoring of Motor RPM enables verification of proper pump operation, assists in identifying abnormal motor behaviour, and provides valuable information during maintenance activities. The simultaneous monitoring of both Engine RPM and Motor RPM provided a comprehensive overview of the operating condition of the coolant conditioning system.

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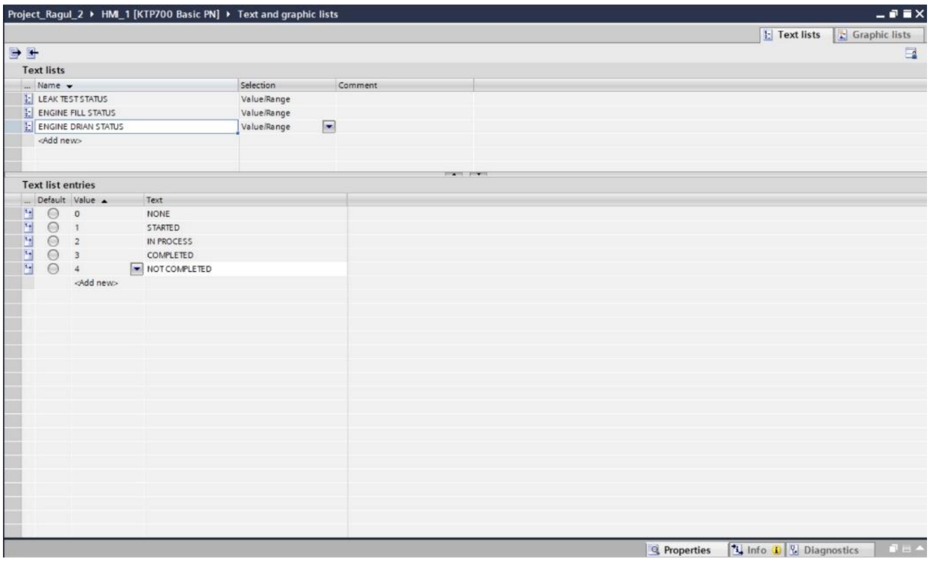
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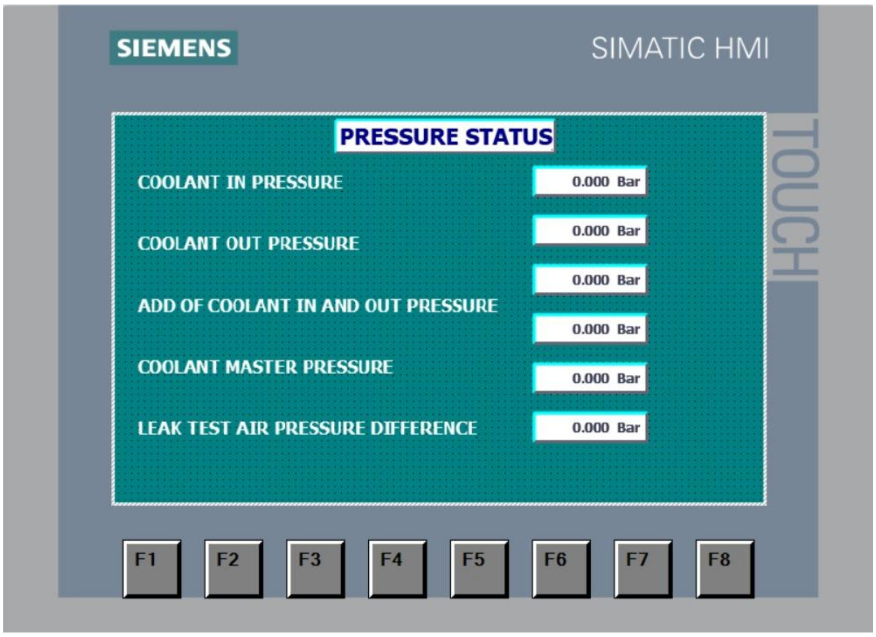
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SIMATIC HMI
TOUCH
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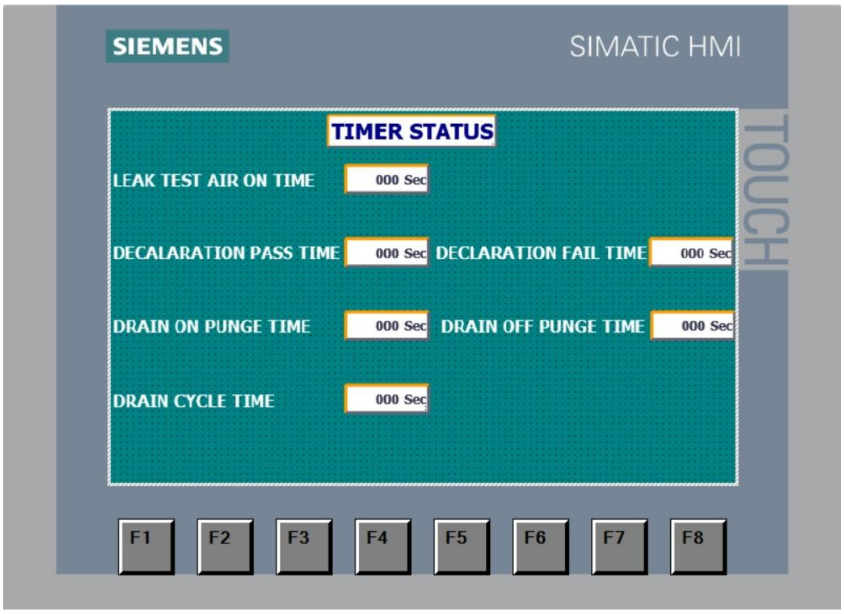


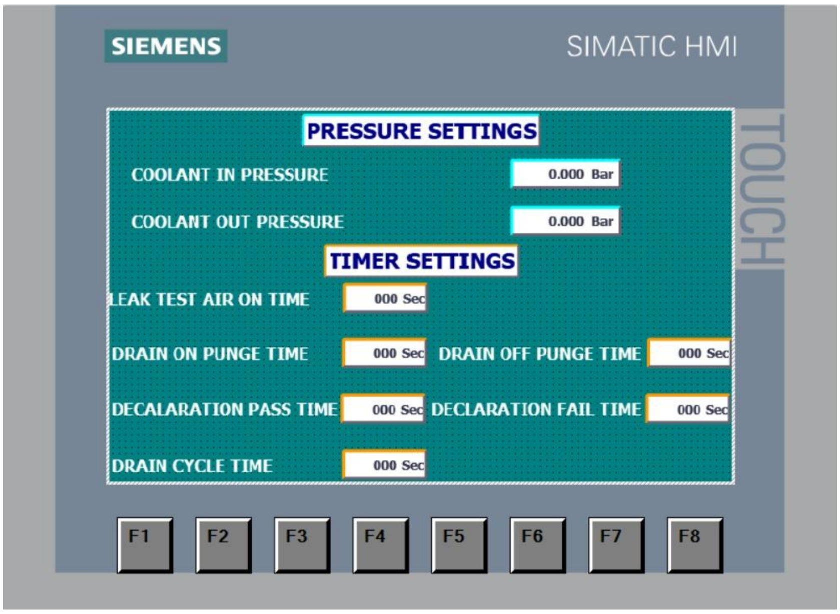


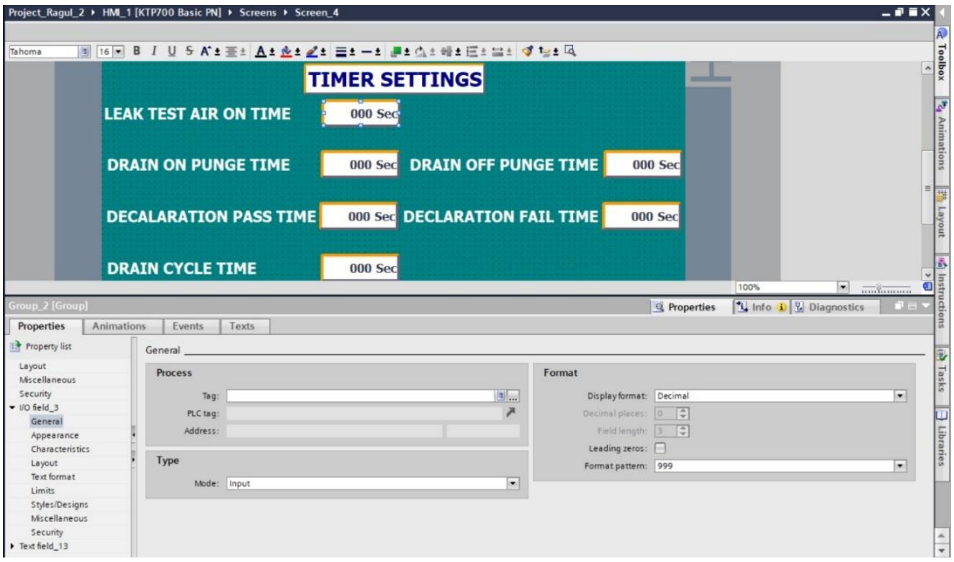


*Figures of the HMI Design and Development*

## TESTING AND VALIDATION OF RPM MEASUREMENT

Following completion of the PLC programming, comprehensive testing was carried out to verify the accuracy and reliability of the developed RPM monitoring system. Initially, individual High-Speed Counter channels were tested independently to verify correct pulse acquisition. The Engine RPM and Motor RPM channels were then operated simultaneously to ensure that both counters functioned correctly without interfering with one another. The Cyclic Interrupt execution was monitored to confirm stable execution at the specified interval, while the calculated RPM values were continuously compared with the expected machine operating conditions. Communication between the PLC and Human Machine Interface was also verified to ensure that the calculated RPM values were updated without noticeable delay. Repeated testing confirmed that the High-Speed Counter implementation, Cyclic Interrupt execution, frequency calculation, and RPM conversion algorithm operated correctly under normal operating conditions. The successful validation demonstrated the capability of the PLC-based monitoring system to provide accurate and reliable real-time speed measurement for the Coolant Conditioning System.

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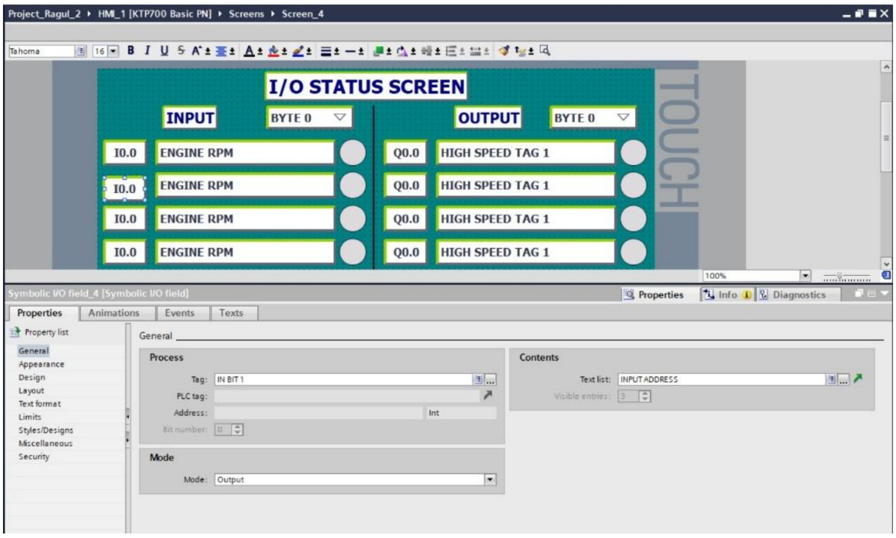


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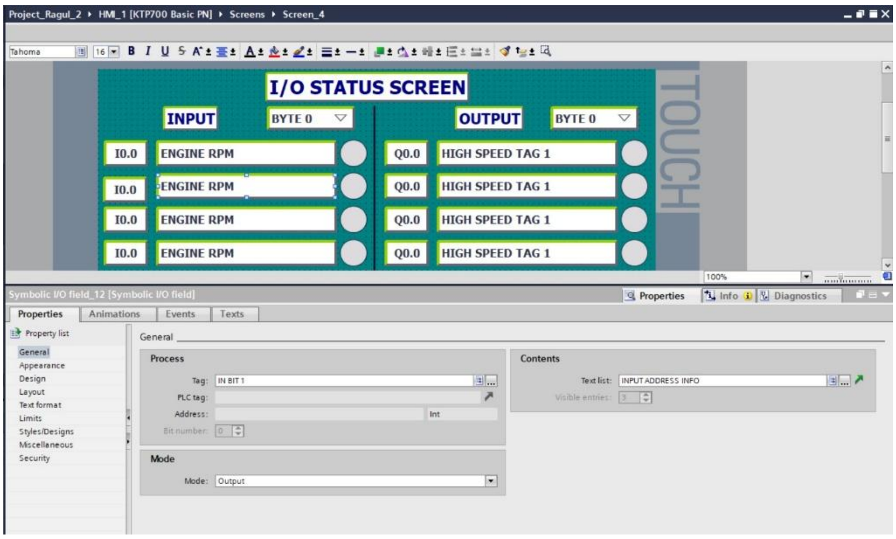

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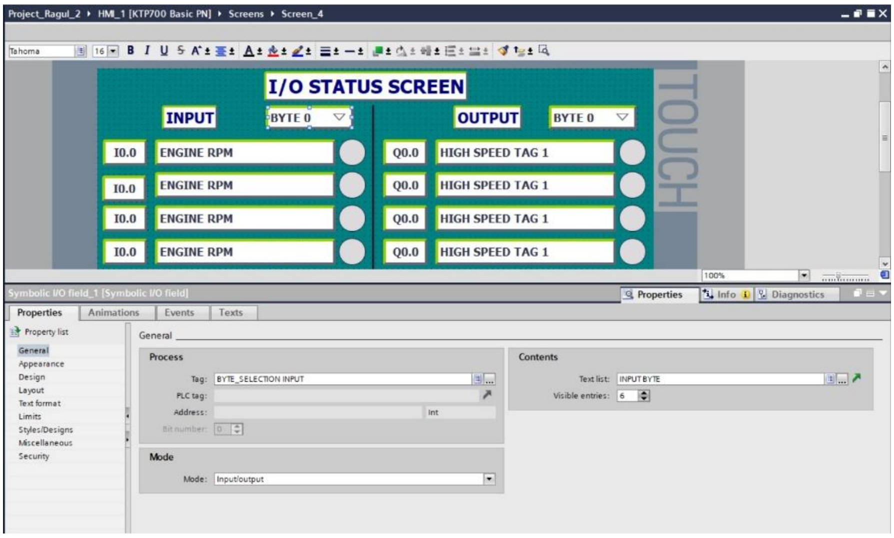

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Info
Diagnostics
Properties
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Events
Texts
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Design
Layout
Text format
Limits
Styles/Designs
Miscellaneous
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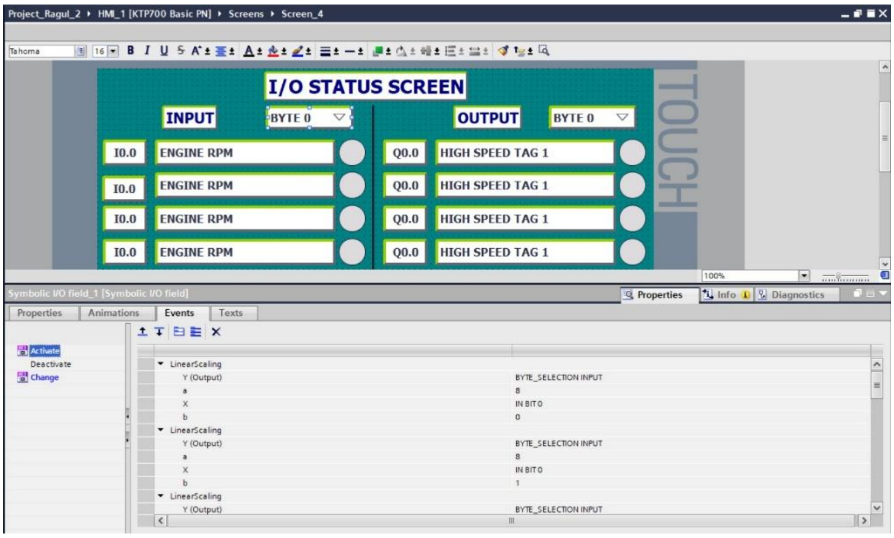

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Q0.0
HIGH SPEED TAG 1
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Symbolic I/O field_1 [Symbolic I/O field]
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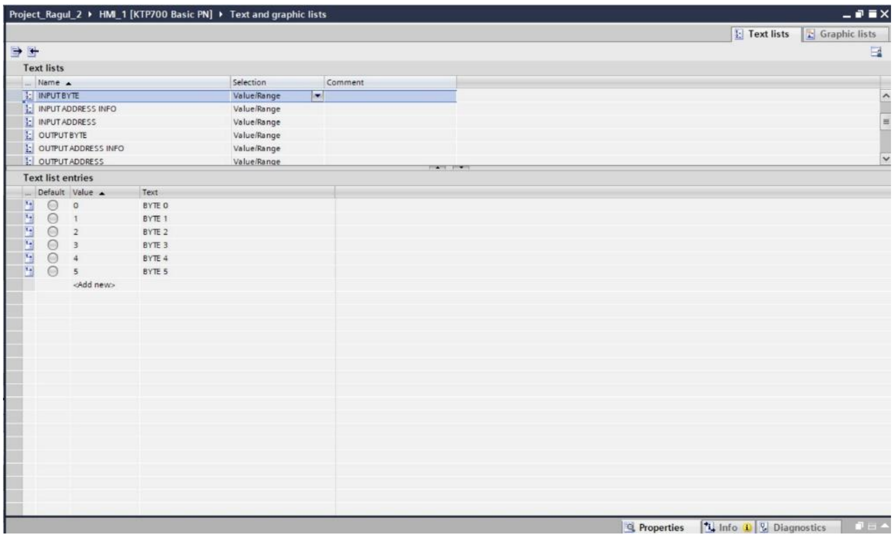

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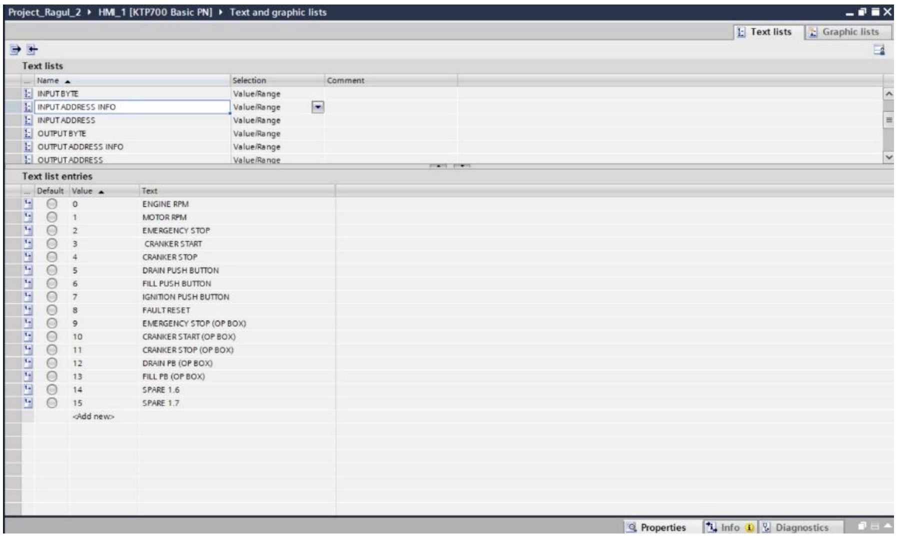
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ValueRange
Text list entries
Default
Value
Text
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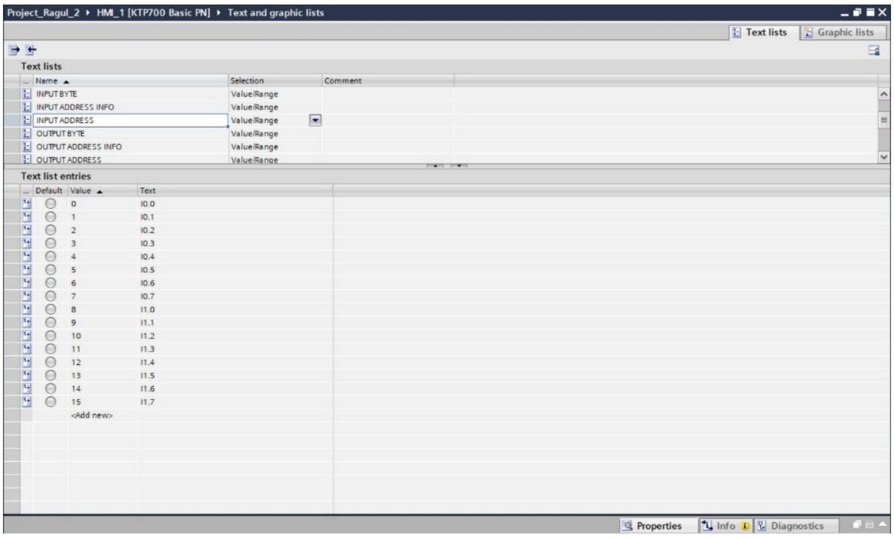
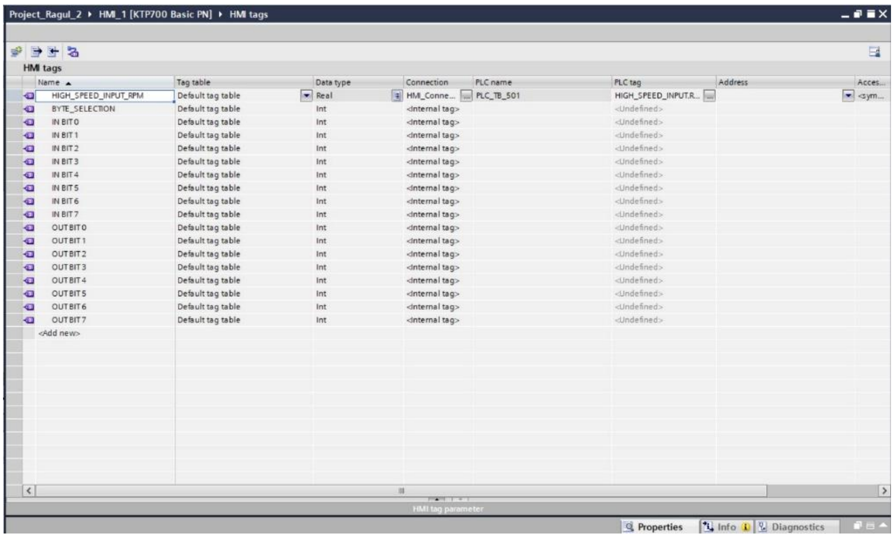
1 MOTOR RPM
2 EMERGENCY STOP
3 CRANKER START
4 CRANKER STOP
5 DRAIN PUSH BUTTON
6 FILL PUSH BUTTON
7 IGNITION PUSH BUTTON
8 FAULT RESET
9 EMERGENCY STOP (OP BOX)
10 CRANKER START (OP BOX)
11 CRANKER STOP (OP BOX)
12 DRAIN PB (OP BOX)
13 FILL PB (OP BOX)
14 SPARE 1.6
15 SPARE 1.7
<Add new>
Properties
Info
Diagnostics

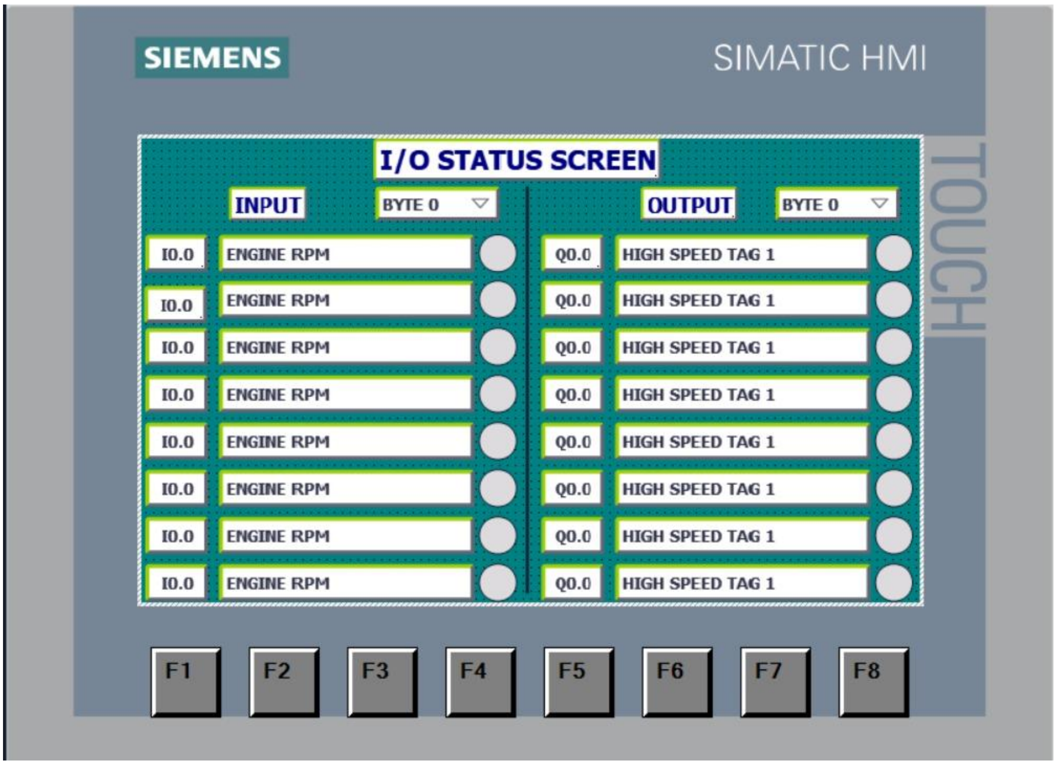
7.


Project_Ragul_2 ▸ HMI_1 [KTP700 Basic PN] ▸ Text and graphic lists
Text lists
Graphic lists
Text lists
Name
Selection
Comment
INPUT BYTE
ValueRange
INPUT ADDRESS INFO
ValueRange
INPUT ADDRESS
ValueRange
OUTPUT BYTE
ValueRange
OUTPUT ADDRESS INFO
ValueRange
OUTPUT ADDRESS
ValueRange
Text list entries
Default
Value
Text
0 I0.0
1 I0.1
2 I0.2
3 I0.3
4 I0.4
5 I0.5
6 I0.6
7 I0.7
8 I1.0
9 I1.1
10 I1.2
11 I1.3
12 I1.4
13 I1.5
14 I1.6
15 I1.7
<Add new>
Properties
Info
Diagnostics

8.


Project_Ragul_2 ▸ HMI_1 [KTP700 Basic PN] ▸ HMI tags
HMI tags
Name
Tag table
Data type
Connection
PLC name
PLC tag
Address
Acces...
HIGH_SPEED_INPUT_RPM Default tag table Real HMI_Conne... PLC_TB_501 HIGH_SPEED_INPUT.R... <sym...
BYTE_SELECTION Default tag table Int <Internal tag> <Undefined>
IN BIT 0 Default tag table Int <Internal tag> <Undefined>
IN BIT 1 Default tag table Int <Internal tag> <Undefined>
IN BIT 2 Default tag table Int <Internal tag> <Undefined>
IN BIT 3 Default tag table Int <Internal tag> <Undefined>
IN BIT 4 Default tag table Int <Internal tag> <Undefined>
IN BIT 5 Default tag table Int <Internal tag> <Undefined>
IN BIT 6 Default tag table Int <Internal tag> <Undefined>
IN BIT 7 Default tag table Int <Internal tag> <Undefined>
OUTBIT 0 Default tag table Int <Internal tag> <Undefined>
OUTBIT 1 Default tag table Int <Internal tag> <Undefined>
OUTBIT 2 Default tag table Int <Internal tag> <Undefined>
OUTBIT 3 Default tag table Int <Internal tag> <Undefined>
OUTBIT 4 Default tag table Int <Internal tag> <Undefined>
OUTBIT 5 Default tag table Int <Internal tag> <Undefined>
OUTBIT 6 Default tag table Int <Internal tag> <Undefined>
OUTBIT 7 Default tag table Int <Internal tag>
<Add new>
HMI tag parameter
Properties
Info
Diagnostics

9.




*Figures of the HMI Design and Development for I/O Status using Bit Indexing Method*

## CHALLENGES ENCOUNTERED DURING THE PROJECT

The development of the PLC-based Coolant Conditioning System involved several technical and practical challenges that required systematic engineering analysis and problem-solving. Since the project involved real-time monitoring of rotating equipment, special attention was required to ensure accurate acquisition and processing of high-speed pulse signals. One of the initial challenges involved understanding the existing coolant conditioning process and identifying the parameters that required monitoring. A detailed study of the system architecture, equipment operation, instrumentation arrangement, and existing control methodology was necessary before developing the automation solution.

The implementation of High-Speed Counter technology presented another significant challenge. Since Engine RPM and Motor RPM signals were generated at high frequencies, it was essential to ensure that pulse acquisition occurred accurately without signal loss. Proper configuration of the High-Speed Counter channels and verification of pulse acquisition were therefore critical stages of the project. Another challenge involved selecting an appropriate execution interval for RPM calculation. The use of a Cyclic Interrupt with an execution time of less than 10 milliseconds required careful program organization to ensure that all calculations were completed within the available execution window while maintaining system stability. The development of the RPM calculation algorithm also required verification of pulse frequency measurements, Pulse Per Revolution (PPR) values, and conversion calculations to ensure accurate speed determination under varying operating conditions. Communication between the PLC and Human Machine Interface required proper tag configuration and data mapping. Every calculated parameter had to be transferred correctly to the HMI while maintaining real-time updates for the operator. Extensive testing and validation activities were therefore carried out throughout the development process to ensure reliable operation of the complete monitoring system.

## RESULTS AND ACHIEVEMENTS

The successful implementation of the PLC-based Coolant Conditioning System resulted in a reliable and effective automation solution for monitoring important operating parameters associated with coolant circulation and engine testing activities. The developed system successfully acquired high-speed pulse signals from both the engine and motor, processed the acquired data using High-Speed Counter technology, calculated the corresponding RPM values, and displayed the information on the Human Machine Interface in real time. The integration of High-Speed Counters enabled accurate speed monitoring without pulse loss, thereby improving measurement reliability compared with conventional PLC input processing methods.

The use of Cyclic Interrupt execution ensured consistent data acquisition and improved responsiveness of the RPM monitoring system. The calculated values remained stable and accurately reflected the operating condition of the rotating equipment. The Human Machine Interface provided operators and maintenance personnel with immediate access to important operating information, thereby improving system visibility and simplifying troubleshooting activities. The project demonstrated the successful application of industrial automation principles to process monitoring and equipment diagnostics within a manufacturing environment.

## ADVANTAGES OF THE DEVELOPED SYSTEM

The PLC-based Coolant Conditioning System offers several operational and maintenance advantages compared with conventional monitoring methods. Some of the major advantages observed during the project are listed below:

- Accurate real-time monitoring of Engine RPM.
- Accurate real-time monitoring of Motor RPM.
- Reliable pulse acquisition using High-Speed Counter technology.
- Elimination of pulse loss during high-speed operation.
- Improved visibility of equipment operating conditions.
- Faster identification of abnormal operating conditions.
- Simplified troubleshooting through PLC diagnostics.
- Continuous monitoring through Human Machine Interface visualization.
- Improved maintenance efficiency through real-time data availability.

- Enhanced system reliability through programmable automation.
- Reduced dependency on manual observation and measurement.
- Improved documentation and monitoring capability.
- Easy future expansion of monitoring functions.
- Improved integration between field devices and automation systems.
- Better utilization of industrial automation technology within the coolant conditioning process.

## FUTURE SCOPE

Although the developed system successfully performs RPM monitoring and process visualization, several advanced automation features can be incorporated in future to further enhance system capability and operational efficiency. Additional process parameters such as coolant temperature, coolant pressure, coolant flow rate, motor current, pump vibration levels, and bearing temperatures can be integrated into the PLC monitoring system. This would provide a more comprehensive understanding of equipment health and process performance.

The system can also be integrated with Supervisory Control and Data Acquisition (SCADA) software to enable centralized monitoring, alarm management, historical data storage, trend analysis, and report generation. Predictive maintenance features can be developed by continuously monitoring equipment behaviour and analysing operating trends. Such systems can provide early warnings regarding potential equipment failures and significantly reduce unplanned downtime.

Remote monitoring through Industrial Ethernet networks can also be implemented, enabling maintenance engineers to access system information from centralized maintenance workstations. Advanced analytics and Industry 4.0 technologies may further be incorporated to improve equipment reliability, process optimization, and intelligent maintenance decision-making. These future enhancements demonstrate the scalability and flexibility of the developed PLC-based monitoring architecture.

## TECHNICAL KNOWLEDGE AND SKILLS ACQUIRED

The Coolant Conditioning System project provided extensive practical exposure to industrial automation, process monitoring, instrumentation, and PLC programming technologies. The major knowledge and skills acquired during the project are summarized below:

- Understanding the working principle of industrial coolant conditioning systems.
- Studying coolant circulation processes used in engine testing facilities.
- Practical exposure to Siemens PLC hardware and industrial automation architecture.
- Configuration and utilization of High-Speed Counter technology.

- Understanding pulse acquisition techniques used in industrial automation.
- Development of PLC programs using Siemens TIA Portal.
- Implementation of Cyclic Interrupt-based data acquisition.
- Application of Double Integer (DINT) data types for high-speed counting operations.
- Development of RPM calculation algorithms using frequency-based measurements.
- Understanding Pulse Per Revolution (PPR) concepts and rotational speed calculations.
- Real-time monitoring of Engine RPM and Motor RPM.
- Human Machine Interface development and configuration.
- PLC-HMI communication and data transfer.
- Industrial instrumentation and sensor integration.
- Testing, debugging, and validation of automation systems.
- Troubleshooting of PLC-based monitoring applications.
- Engineering documentation and project execution practices.
- Understanding industrial process monitoring and maintenance applications.

# OVERALL INTERNSHIP CONCLUSION AND LEARNING OUTCOMES

The internship at Ashok Leyland Limited, Hosur – I provided invaluable industrial exposure and significantly enhanced technical knowledge, practical skills, and professional competency. The overall learning outcomes of the internship are summarized below:

- Gained comprehensive exposure to the manufacturing environment of one of India's leading commercial vehicle manufacturers.
- Developed a thorough understanding of engine remanufacturing processes carried out in the ALRECON Department.
- Learned the complete workflow involved in engine dismantling, inspection, machining, reconditioning, assembly, testing, and dispatch.
- Understood the importance of dimensional inspection, quality control, and standard operating procedures in ensuring product reliability.
- Acquired practical knowledge of engine performance testing using a dynamometer and the evaluation of engine operating parameters.
- Gained an understanding of inventory management, production planning, and material traceability practices followed in industrial manufacturing.
- Observed the implementation of Lean Manufacturing concepts, including 5S workplace organization and the Ashok Leyland Production System (ALPS).
- Developed a clear understanding of preventive, predictive, and breakdown maintenance practices followed in an automobile manufacturing industry.
- Learned the importance of maintenance planning in minimizing equipment downtime and improving production efficiency.
- Gained practical exposure to industrial electrical systems, control panels, motors, contactors, relays, circuit breakers, and protection devices.
- Developed knowledge of industrial sensors, actuators, and instrumentation used in automated manufacturing systems.
- Acquired practical understanding of Programmable Logic Controllers (PLC) and their applications in industrial automation.
- Gained hands-on exposure to Siemens Totally Integrated Automation (TIA) Portal for PLC programming and system configuration.
- Learned the principles of Human Machine Interface (HMI) development for industrial process monitoring and operator interaction.
- Understood industrial communication between PLCs, HMIs, and field devices within automation systems.

- Participated in the design and development of a PLC-based Air Handling Unit (AHU) control panel for Shop Floor 5.
- Gained practical experience in electrical panel design, component selection, wiring practices, PLC programming, HMI development, testing, and system verification.
- Developed an understanding of industrial HVAC systems and manual speed selection control strategies using PLCs.
- Participated in the development of a PLC-based Coolant Conditioning System for Shop Floor 5.
- Learned the working principles of coolant circulation systems used in engine testing applications.
- Acquired practical knowledge of High-Speed Counter (HSC) technology for processing high-frequency pulse signals.
- Implemented Engine RPM and Motor RPM monitoring using High-Speed Counter channels ID1000 and ID1004.
- Learned the application of Cyclic Interrupt programming for deterministic execution of real-time calculations.
- Understood the methodology for calculating rotational speed using frequency measurement and Pulse Per Revolution (PPR).
- Gained practical experience in using the Double Integer (DINT) data type for high-speed industrial calculations.
- Improved knowledge of industrial troubleshooting techniques through testing, debugging, and system validation activities.
- Strengthened analytical thinking and logical problem-solving abilities while working on industrial automation projects.
- Developed the ability to analyse engineering problems and identify suitable technical solutions.
- Improved technical documentation, report writing, and engineering presentation skills through project documentation.
- Enhanced teamwork and communication skills by working with maintenance engineers, supervisors, technicians, and production personnel.
- Understood the importance of industrial safety practices, workplace discipline, and adherence to standard operating procedures.
- Developed professional work ethics, punctuality, responsibility, and effective time management throughout the internship.
- Gained confidence in applying theoretical Production Engineering concepts to practical

industrial applications.

- Acquired multidisciplinary knowledge integrating mechanical, electrical, electronics, instrumentation, and automation engineering principles.
- Strengthened practical understanding of modern manufacturing systems and Industry 4.0-oriented automation technologies.
- Improved readiness to undertake future responsibilities in production engineering, industrial automation, manufacturing systems, maintenance engineering, robotics, and intelligent manufacturing.
- Successfully bridged the gap between academic learning and real industrial practice through active participation in manufacturing operations, maintenance activities, and automation projects.
- Overall, the internship served as a highly enriching learning experience that significantly enhanced technical competence, engineering knowledge, practical skills, analytical ability, professional confidence, and overall career preparedness as a Production Engineering student.